

OG-RAG: 本体引导检索增强生成用于大语言模型

预印本

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摘要

本文介绍了OG-RAG，这是一种本体引导检索增强生成方法，旨在通过将检索过程锚定在领域特定本体中来增强LLM生成响应。虽然大语言模型被广泛用于问答和搜索等任务，但它们在昂贵的微调或次优检索方法的情况下难以适应专业知识，例如工业工作流或知识工作。现有的检索增强模型（如RAG）提供了改进，但它们无法考虑结构化的领域知识，导致上下文生成次优。本体从概念上通过定义实体及其相互关系来组织领域知识，为解决这一差距提供了结构化表示。OG-RAG构建领域文档的超图表示，其中每个超边封装了使用领域特定本体锚定的集群化事实知识。然后，优化算法检索构建精确、概念上锚定的上下文的最小超边集，以供LLM使用。该方法实现了高效检索，同时保留了实体之间的复杂关系。OG-RAG适用于需要基于事实的推理的领域，特别是在需要工作流或决策步骤遵循预定义规则和程序的任务中。这些包括医疗保健、法律和农业部门的工业工作流，以及新闻新闻业、调查研究、咨询等知识驱动型任务。我们的评估表明，OG-RAG将准确事实的召回率提高了55%，并将四个不同LLM的响应正确性提高了40%。此外，与基线方法相比，OG-RAG将响应到上下文的归因速度提高了30%，并将基于事实的推理准确率提高了27%。

关键词 检索增强生成，知识图谱，事实演绎，上下文归因，本体接地检索

1 引言

大语言模型（LLMs）提升了问答系统、搜索引擎和任务导向型聊天机器人的能力 [Perplexity, 2024, ChatGPT, 2024, Achiam等人, 2023]。然而，它们在基于事实的适应性方面面临重大挑战，特别是在依赖精确、领域特定数据的领域 [Cascella等人, 2023, Thirunavukarasu等人, 2023, Singhal等人, 2023, Guha等人, 2024, Wang等人, 2024, Balaguer等人, 2024]。考虑一个精准农业系统，其中土壤湿度和天气数据的实时变化必须影响灌溉决策。一个通用型LLM可能会根据广泛知识提出灌溉计划，但无法考虑该地区的特定土壤条件或植物需求。这种缺乏适应性意味着LLM的建议可能不准确，可能导致过度浇水或灌溉不足，从而损害作物。此类场景突显了一个核心限制：LLM无法可靠地适应领域特定决策，而准确性和专业知识至关重要 [Rudin, 2019, Sharma等人, 2020]。

为了克服这些局限性，现成的LLM可以针对特定领域进行微调 [Bommasani等人2021]，或者与外部工具或文档结合使用 [Lewis等人, 2020年, 庄等人, 2023年, Schick等人 2024]。

OG-RAG: ONTOLOGY-GROUNDED RETRIEVAL-AUGMENTED
GENERATION FOR LARGE LANGUAGE MODELS

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ABSTRACT

This paper presents OG-RAG, an Ontology-Grounded Retrieval Augmented Generation method designed to enhance LLM-generated responses by anchoring retrieval processes in domain-specific ontologies. While LLMs are widely used for tasks like question answering and search, they struggle to adapt to specialized knowledge, such as industrial workflows or knowledge work, without expensive fine-tuning or sub-optimal retrieval methods. Existing retrieval-augmented models, such as RAG, offer improvements but fail to account for structured domain knowledge, leading to suboptimal context generation. Ontologies, which conceptually organize domain knowledge by defining entities and their interrelationships, offer a structured representation to address this gap. OG-RAG constructs a hypergraph representation of domain documents, where each hyperedge encapsulates clusters of factual knowledge grounded using domain-specific ontology. An optimization algorithm then retrieves the minimal set of hyperedges that constructs a precise, conceptually grounded context for the LLM. This method enables efficient retrieval while preserving the complex relationships between entities. OG-RAG applies to domains where fact-based reasoning is essential, particularly in tasks that require workflows or decision-making steps to follow predefined rules and procedures. These include industrial workflows in healthcare, legal, and agricultural sectors, as well as knowledge-driven tasks such as news journalism, investigative research, consulting and more. Our evaluations demonstrate that OG-RAG increases the recall of accurate facts by 55% and improves response correctness by 40% across four different LLMs. Additionally, OG-RAG enables 30% faster attribution of responses to context and boosts fact-based reasoning accuracy by 27% compared to baseline methods.

Keywords Retrieval Augmented Generation, Knowledge Graphs, Factual Deduction, Context Attribution, Ontology Grounded Retrieval

1 Introduction

Large language models (LLMs) have advanced the capabilities of question-answering systems, search engines, and task-oriented chatbots [Perplexity, 2024, ChatGPT, 2024, Achiam et al., 2023]. However, they face significant challenges with fact-based adaptation, particularly in domains that rely on precise, domain-specific data [Cascella et al., 2023, Thirunavukarasu et al., 2023, Singhal et al., 2023, Guha et al., 2024, Wang et al., 2024, Balaguer et al., 2024]. Consider a precision agriculture system where real-time changes in soil moisture and weather data must influence irrigation decisions. A general-purpose LLM might suggest irrigation plans based on broad knowledge but fail to account for specific soil conditions or plant requirements in that region. This lack of adaptability means the LLM’s recommendation could be inaccurate, potentially leading to overwatering or under-irrigation, which can harm crops. Such scenarios highlight a core limitation: the inability of LLMs to reliably adapt to domain-specific decision-making, where accuracy and specialized knowledge are paramount [Rudin, 2019, Sharma et al., 2020].

To overcome these limitations, off-the-shelf LLMs can be either fine-tuned for specific domains [Bommasani et al., 2021] or paired with external tools or documents [Lewis et al., 2020, Zhuang et al., 2023, Schick et al., 2024].

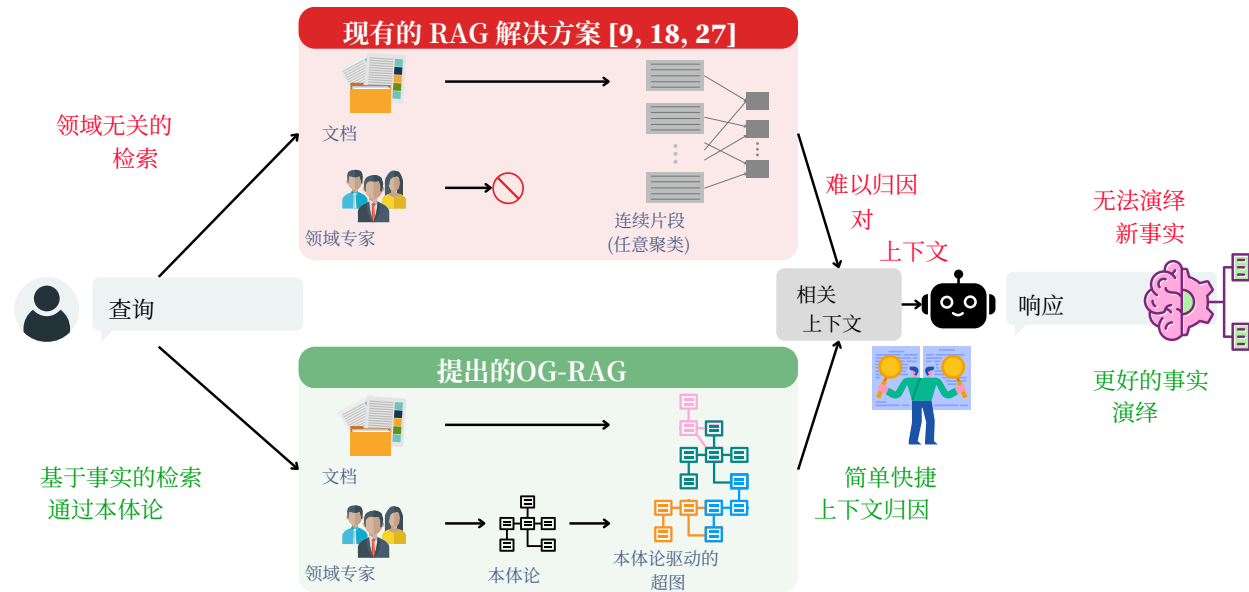


图1: 提出的OG-RAG与现有检索增强生成 (RAG) 解决方案的比较。

然而，微调计算成本高且需要大量数据管理，使其成为一种不太实用的解决方案 [Balaguer等人, 2024年, Ovadia等人 2023]。另一方面，基于检索的方法，例如RAG Lewis等人 [2020], Sarthi等人 [2024], 张等人 [2024], Borgeaud等人 [2022], Karpukhin等人 [2020], Edge等人[2024], 使用领域无关的嵌入来从特定领域的文档中检索与查询相关的信息，并使用检索到的信息来回答。尽管有前景，但这些方法无法捕捉到准确领域特定检索所需的深层概念关系和细微事实。

每个领域都以独特的方式组织其知识和术语，这些方式无法跨不同领域进行泛化 [Mernik等人, 2005]。例如，在工业工作流程中，事实和关系被仔细策划并结构化为领域特定的框架，而在知识工作和调查研究领域，本体论作为组织和分析事实和概念的模板 [Jackson, 1990, Guarino等人, 2009]。当前的大语言模型难以适应这些多样化的结构，限制了它们在专业领域中的准确性和有效性。另一个主要问题是，用户往往难以将生成的响应追溯到相关上下文。此外，许多专业领域遵循严格的程序规则，而当前技术无法可靠地根据已建立的领域知识推导出准确的结论。这一差距对大语言模型在专业工作流程中的更广泛应用构成了重大挑战。

在本文中，我们提出了OG-RAG（本体论-基础检索增强生成），通过集成领域特定本体论来弥补现有大语言模型的这一差距。本体论定义了领域内关键实体及其关系，提供了结构化表示，这对于适应复杂和不断变化的信息环境至关重要。如图1所示，OG-RAG利用这些本体论通过在结构化领域知识内进行检索来增强大语言模型响应，从而提高响应准确性，支持基于事实的灵活适应，并实现可验证的上下文归因。

OG-RAG的一个关键特性是它使用领域文档的超图表示，这是一种比传统检索方法更复杂和多维的方式来建模关系。超图中的每个超边代表基于相应本体论的一组相关事实知识。使用贪婪算法，引擎检索给定查询的最小超边集合，为大型语言模型形成一个紧凑的上下文。与传统的检索方法不同，这种方法有效地将复杂关系和领域特定知识提炼为结构化上下文，使大型语言模型能够生成上下文感知的响应，而不会增加显著的计算开销。

OG-RAG 适用于需要基于事实进行适配的广泛领域。这些领域包括医疗保健、法律和农业部门的工业工作流程，以及新闻业、基于网络的调查研究、咨询等知识型工作。我们在农业和新闻领域的 OG-RAG 评估表明，OG-RAG 将准确事实的召回率提高了 55%，并在四个不同的大语言模型上，将生成响应的整体正确性提升了 40%。我们的用户研究表明，将大语言模型的响应归因于 OG-RAG 检索的上下文比其他方法快 30% 且效果更好。最后，在基于事实的推理任务中，我们发现与其他方法相比，应用预定义规则于 OG-RAG 的上下文时，大语言模型的响应正确率提高了 27%。

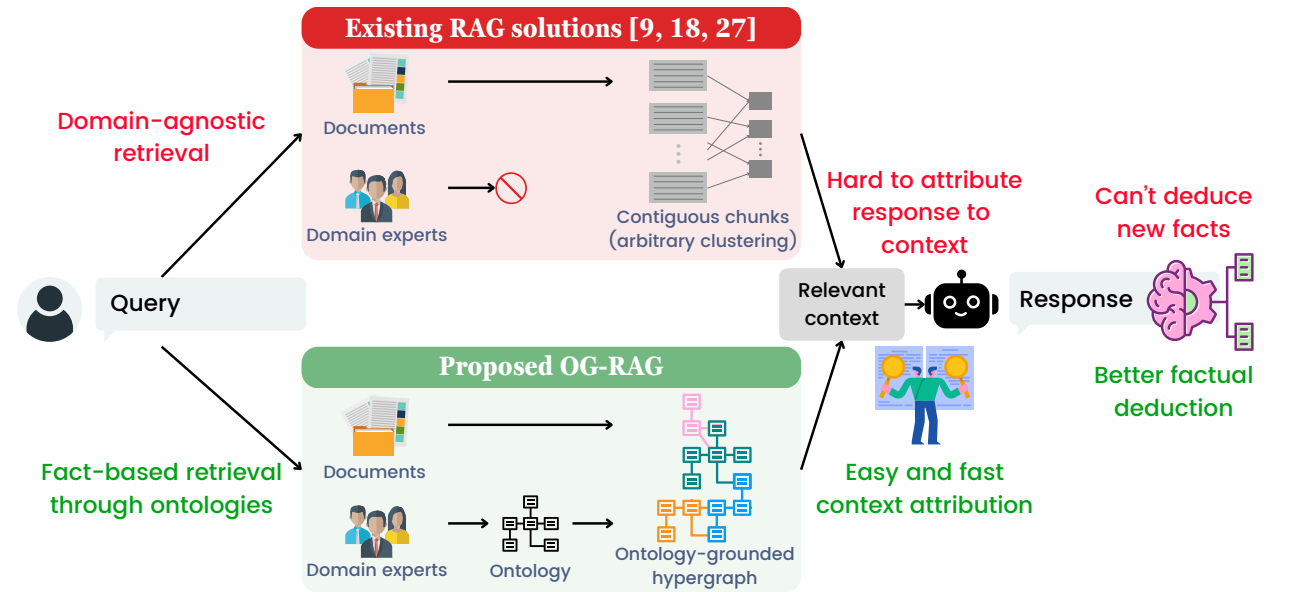


Figure 1: Comparison of the proposed OG-RAG with existing retrieval-augmented generation (RAG) solutions.

However, fine-tuning is computationally expensive and requires extensive data curation, making it a less practical solution [Balaguer et al., 2024, Ovadia et al., 2023]. On the other hand, retrieval-based approaches, such as RAG Lewis et al. [2020], Sarthi et al. [2024], Zhang et al. [2024], Borgeaud et al. [2022], Karpukhin et al. [2020], Edge et al. [2024], use domain-agnostic embeddings to retrieve query-relevant information from domain-specific documents and use the retrieved information for answering. Although promising, these methods fail to capture the deep conceptual relationships and nuanced facts required for accurate domain-specific retrieval.

Each domain organizes its knowledge and terminology in distinct ways, which cannot be generalized across different fields [Mernik et al., 2005]. For example, in industrial workflows, facts and relationships are carefully curated and structured into domain-specific frameworks, while in knowledge work and investigative research, ontologies serve as templates for organizing and analyzing facts and concepts [Jackson, 1990, Guarino et al., 2009]. Current LLMs struggle to adapt to these diverse structures, limiting their accuracy and effectiveness in specialized domains. Another major issue is that users often struggle to trace generated responses back to the relevant context. Furthermore, many specialized domains follow strict procedural rules, and the current techniques fail to reliably deduce accurate conclusions based on this established domain knowledge. This gap presents a major challenge to the wider applicability of LLMs in specialized workflows.

In this paper, we present OG-RAG (Ontology-Grounded Retrieval Augmented Generation) that bridges this gap of existing LLMs by integrating domain-specific ontologies for fact-based adaptation. Ontologies, which define key entities and their relationships within a domain, provide structured representation that is essential for adapting to complex and evolving information landscapes. As shown in Figure 1, OG-RAG leverages these ontologies to enhance LLM responses by grounding retrieval within structured domain knowledge, leading to *improved response accuracy*, supporting *flexible fact-based adaptation*, and enabling *verifiable context attribution*. A key feature of OG-RAG is its use of hypergraph representations of domain documents, which is a more sophisticated and multi-faceted way to model relationships than traditional retrieval approaches. Each hyperedge in the hypergraph represents a collection of related factual knowledge based on the corresponding ontology. Using a greedy algorithm, the engine retrieves a minimal set of hyperedges for a given query that forms a compact context for the LLM. Unlike traditional retrieval approaches, this method effectively distills complex relationships and domain-specific knowledge into a structured context, adapting LLMs to generate context-aware responses without adding significant computational overhead.

OG-RAG applies to a wide set of domains where fact-based adaptation is essential. These include industrial workflows in healthcare, legal, and agricultural sectors, as well as knowledge work such as news journalism, web based investigative research, consulting, and more. Our evaluations of OG-RAG within the agriculture and news domains demonstrate that OG-RAG increases the recall of accurate facts by 55%, and improves the overall correctness of generated responses by 40% across four different LLMs. Our user study shows that attributing LLM responses to the context retrieved by OG-RAG is 30% faster and better. Finally, in a fact-based reasoning task, we find that LLM responses are 27% more

这些结果突显了 OG-RAG 在专业工作流中提供更可靠、基于事实的答案方面的有效性。

2 相关工作

微调。一种克服大语言模型局限性的方法是使用领域特定数据 [Bommasani等人, 2021]进行微调。微调允许模型通过在专门数据集上重新训练来适应特定领域的细微差别。然而, 这种方法计算成本高昂, 需要大量资源和广泛的数据管理, 这使得它在许多实际应用中不切实际 [Kumar等人, 2022]。OG-RAG通过基于检索的解决方案消除了对昂贵微调的需求, 从而解决了这一缺点。

幻觉缓解。大语言模型容易产生幻觉, 即, 与输入无关或事实性错误输出 [Ji等人, 2023a]。这些幻觉在需要精确度的领域 (如科学研究或工业工作流) 中尤其成问题 [Thirunavukarasu等人, 2023]。现有系统已尝试通过生成后校正方法和事实性检查来缓解幻觉, 但这些方法通常需要额外的计算层, 且并非万无一失 [Ji等人, 2023b, Madaan等人, 2024, Welleck等人, 2024]。OG-RAG通过将数据映射本体转换为超图, 并使用优化的相关事实簇检索来减少幻觉, 确保大语言模型响应基于领域特定事实。

检索方法除了传统的检索增强生成 (RAG) [Lewis等人, 2020], 还提出了基于图的检索方法。这些方法包括 GraphRAG [Edge等人, 2024], RAPTOR [Sarathi等人, 2024],以及其他基于知识图谱的框架, 如 Langchain [lan] 和 Neo4J [neo]。它们通过利用结构化知识图谱来组织和检索上下文相关信息, 提升了大语言模型 (LLM) 的性能。GraphRAG通过组织实体和关系进行语义聚类, 能够更高效地处理复杂查询, 而RAPTOR采用分层结构进行多级抽象, 提高了对大型文档的上下文理解能力。然而, 这些方法依赖于针对实体和领域特定信息的临时提取, 往往缺乏领域专业知识的基础。这导致生成正确结构化表示的工作流程过于复杂, 同时仍然存在显著的精度差距。此外, 这也导致了更弱的上下文归因, 使得将结论追溯到相关事实变得更加困难。在现有方法的基础上, OG-RAG的超边构建提供了紧凑的事实表示, 通过更好的上下文归因增强了透明度, 而其超图检索机制则选择与查询精确匹配的最佳事实簇。

归属。为了增强大语言模型响应的可解释性和可靠性, 将其生成归因于可信来源非常重要。一种方法是通过生成带引用的文本, 但已有研究表明现有零样本方法的局限性 [Gao 等人, 2023] 以及专门训练的模型 [Khalifa 等人, 2024]。此外, 还探索了其他形式的归属, 因为引用要求用户在整页上搜索以验证生成响应中的声明, 这并不理想。因此, 局部可归属方法 [Slobodkin 等人, 2024]以及人机协作 Kamalloo 等人。[2023] 策略也已提出。虽然这些方法提供句子级别的归属, 但通过可解释的 RAG 上下文可以实现补充效益。OG-RAG 提供易于归属的上下文, 用户只需少量努力即可追踪响应的生成。

演绎推理。传统的基于规则的推理系统提供了可解释且易于控制的方法, 从给定输入中演绎出新颖的结论 [Jackson, 1990, Saparov 等人, 2023]。然而, 它们缺乏神经模型 (如大语言模型) 的灵活性和泛化能力。另一方面, 大语言模型在演绎推理中容易产生任意幻觉, 这在结构化工作流程中可能存在问题 [Wang 等人, 2024, Saparov 等人, 2023]。OG-RAG 通过将非结构化文本锚定到特定领域的词汇, 结合了基于事实推理的结构化精度和神经灵活性, 使大语言模型能够更有效地应用特定领域的规则, 同时保持跨多个领域的可扩展性。

3 主要贡献总结

本文提出了几项关键贡献, 旨在解决基于事实的推理和幻觉减少方面的挑战, 以及大语言模型 (LLMs) 中的减少问题:

- **基于事实的上下文检索实现领域自适应:** OG-RAG 是一个集成领域特定本体到 LLM 检索过程的新型框架, 促进跨领域的精确、基于事实的自适应。OG-RAG 通过两个核心机制实现这一点: (a) 通过将数据映射本体转换为超图来形式化事实, 以及 (b) 使用基于优化的超图检索来提取与领域特定查询精确对齐的事实紧凑集群, 从而产生更上下文准确的响应。

correct when applying pre-defined rules over OG-RAG’s context compared to other methods. These results highlight OG-RAG’s effectiveness in providing more reliable, fact-based answers in specialized workflows.

2 Related Work

Fine-tuning. One approach to overcome the limitations of LLMs is fine-tuning on domain-specific data [Bommasani et al., 2021]. Fine-tuning allows models to adapt to the nuances of a specific domain by retraining the model on specialized datasets. However, this method is computationally expensive, requiring significant resources and extensive data curation, which makes it impractical for many real-world applications [Kumar et al., 2022]. OG-RAG addresses this shortcoming by eliminating the need for costly fine-tuning through retrieval-based solutions.

Hallucination mitigation. LLMs are prone to generating hallucinations, *i.e.*, outputs that are factually incorrect or irrelevant to the input [Ji et al., 2023a]. These hallucinations are especially problematic in domains that require precision, such as scientific research or industrial workflows [Thirunavukarasu et al., 2023]. Existing systems have attempted to mitigate hallucinations through post-generation correction methods and factuality checks, but these often require additional layers of computation and are not foolproof [Ji et al., 2023b, Madaan et al., 2024, Welleck et al., 2024]. OG-RAG reduces hallucinations by transforming data-mapped ontologies into hypergraphs and uses optimized retrieval of relevant fact clusters, ensuring LLM responses are grounded in domain-specific facts.

Retrieval methods In addition to the traditional retrieval augmented generation (RAG) [Lewis et al., 2020], graph-based approaches have also been proposed. These include GraphRAG [Edge et al., 2024], RAPTOR [Sarathi et al., 2024], and other knowledge graph-based frameworks such as Langchain [lan] and Neo4J [neo]. They have advanced LLM performance by leveraging structured knowledge graphs to organize and retrieve contextually relevant information. GraphRAG excels in semantic clustering by organizing entities and relationships, allowing for more efficient handling of complex queries, while RAPTOR uses a hierarchical structure for multi-level abstraction, improving contextual understanding across large documents. However, these approaches rely on ad-hoc extraction of entities and domain-specific information, often without grounding in domain expertise. This results in overly complex workflows for generating the correct structured representation, while still leaving significant gaps in precision. It also leads to weaker context attribution, making it more difficult to trace conclusions back to relevant facts. Improving on these approaches, OG-RAG’s hyperedge construction offers a compact fact representation that enhances transparency through better context attribution, while its hypergraph retrieval mechanism selects optimal fact clusters precisely tailored to the query.

Attribution. To enhance the interpretability and reliability of the LLM responses, it is important to attribute their generation to trustworthy sources. One way is to generate text with citations but prior work has shown limitations of existing zero-shot approaches [Gao et al., 2023] and specially-trained models [Khalifa et al., 2024]. Furthermore, other forms of attribution are also explored since citations require users to search over a full page to verify the claims in the generated response, which is undesirable. Thus, locally-attributable methods [Slobodkin et al., 2024] and human-in-the-loop Kamalloo et al. [2023] strategies have also been proposed. While these approaches provide sentence-level attribution, complementary benefits can be achieved through interpretable RAG contexts. OG-RAG provides easy-to-attribute contexts that require only a little effort from the users to trace the generation of the response.

Deductive reasoning. Traditional rule-based reasoning systems provide interpretable and easily controllable ways to deduce novel conclusions from a given input [Jackson, 1990, Saparov et al., 2023]. However, they lack the flexibility and generalization capabilities of neural models like LLMs. On the other hand, LLMs are prone to arbitrary hallucinations in deductive reasoning, which can be problematic in structured workflows [Wang et al., 2024, Saparov et al., 2023]. OG-RAG combines the structured precision of fact-based reasoning with neural flexibility by anchoring unstructured text to domain-specific vocabulary, enabling LLMs to more effectively apply domain-specific rules while maintaining scalability across multiple domains.

3 Summary of Key Contributions

This paper presents several key contributions that address the challenges of fact-based reasoning and hallucination reduction in large language models (LLMs):

- **Fact-based Context Retrieval Enabling Domain Adaptation:** OG-RAG is a novel framework that integrates domain-specific ontologies into the LLM retrieval process, facilitating precise, fact-based adaptation across domains. OG-RAG achieves this through two core mechanisms: (a) formalization of *facts* by transforming data-mapped ontologies into a hypergraph, and (b) using an optimization-based hypergraph retrieval to extract compact clusters of facts that precisely align with domain-specific queries, resulting in more contextually accurate responses.

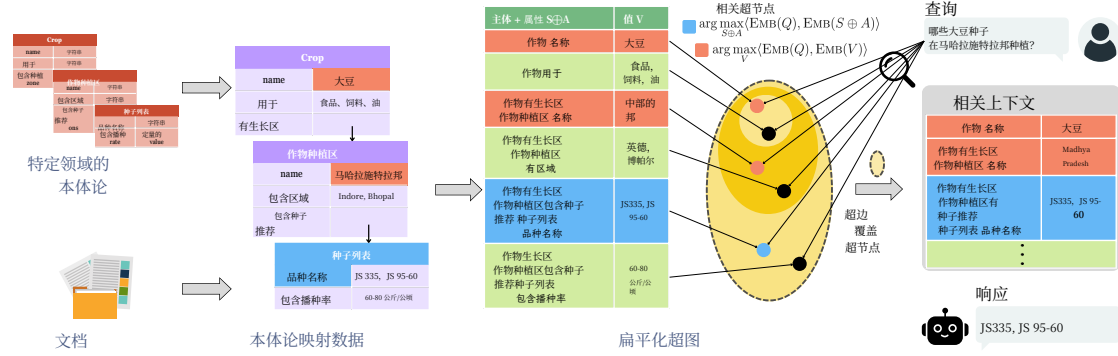


图 2: OG-RAG: 本体引导检索增强生成

- **事实演绎:** OG-RAG 通过利用领域适配的事实来推断结论，从而实现事实演绎。这使系统能够动态生成准确、特定领域的结论，改进在复杂事实环境中的推理，并提高大语言模型（LLM）响应的精确度。
- **改进的上下文归因:** OG-RAG通过提供精确、基于事实的上下文归因而增强响应验证。这使得人类和机器都能够准确地将结论追溯到其底层事实，确保在领域特定查询中进行可靠和透明的fact-checking。

第六节展示了广泛的评估，以支持所列举的每一项贡献。

4 OG-RAG

本体接地检索增强生成（OG-RAG）是一种新型框架，它将本体论——领域特定概念的正式表示及其关系的表示——集成到检索过程中。与现有的检索增强生成（RAG）系统或其他基于本体的方法不同，后者依赖通用嵌入或临时上下文生成，且缺乏领域专业知识的基础，OG-RAG利用本体驱动的超图检索，动态适应结构化知识库和复杂的领域特定查询。图2展示了所提出方法的高级流程，我们在下文将详细描述每个组件。

4.1 超图构建

第一部分包括将通用的领域特定文档 $\mathcal{D}$ 映射到给定的本体论 $\mathcal{O}$ ，并将可用信息转换为检索的可用格式。

4.1.1 本体论

不同的领域通过遵循特定于其独特工作流程的规则和程序来组织决策。例如，在农业中，事实信息的组织方式取决于任务。作物栽培依赖于土壤质量、天气模式和害虫管理策略等事实，其中决策基于表型和环境数据遵循一系列步骤。另一方面，农业预算使用相同的基础事实——例如农场规模或作物类型——但通过财务模型和成本预测来应用它们，这需要不同的决策框架。

一个<样式 id='1'>本体论</样式>是领域内关键实体及其关系的正式表示。例如，在农业领域，定义了作物、土壤和天气条件等实体，以及“作物在某个地区生长”或“土壤具有湿度水平”等关系。通过定义这些实体和关系，本体论为组织领域知识提供了连贯且清晰的框架 [Guarino等人, 2009, Jackson, 1990]。它与<样式 id='5'>分类学</样式>或<样式 id='7'>分类</样式>不同，因为它允许实体之间存在更丰富的、不必是层次结构的关系。更正式地说，

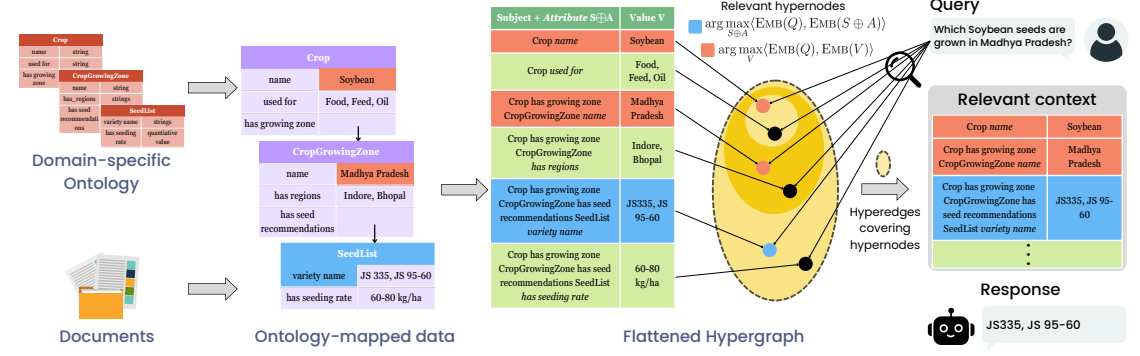


Figure 2: OG-RAG: Ontology-Grounded Retrieval-Augmented Generation

- **Factual Deduction:** OG-RAG enables factual deduction by leveraging domain-adapted facts to infer conclusions based on established knowledge and relationships. This enables the system to generate accurate, domain-specific conclusions dynamically, improving reasoning over complex factual contexts and enhancing the precision of LLM responses.
- **Improved Context Attribution:** OG-RAG enhances response verification by providing precise, fact-based context attribution. This enables both humans and machines to accurately trace conclusions back to their underlying facts, ensuring reliable and transparent fact-checking in domain-specific queries.

Section 6 presents extensive evaluations to support each of the enumerated contributions.

4 OG-RAG

Ontology Grounded Retrieval Augmented Generation (OG-RAG) is a novel framework which integrates ontologies—formal representations of domain-specific concepts and their relationships—into the retrieval process. Unlike existing retrieval-augmented generation (RAG) systems or other ontology based approaches, which rely on general-purpose embeddings or ad-hoc context generation without grounding in domain expertise, OG-RAG leverages ontology-driven hypergraph retrieval to dynamically adapt LLMs to structured knowledge bases and complex domain-specific queries. Figure 2 shows the high-level pipeline of the proposed method while we describe each component in more detail below.

4.1 Hypergraph Construction

The first part includes mapping the general domain-specific documents $\mathcal{D}$ onto a given ontology $\mathcal{O}$ and converting the available information into a usable format for retrieval.

4.1.1 Ontology

Different domains organize decision-making by following specific rules and procedures tailored to their unique workflows. In agriculture, for instance, factual information is organized differently depending on the task. Crop cultivation relies on facts like soil quality, weather patterns, and pest management strategies, where decisions follow a series of steps based on phenotype and environmental data. On the other hand, agricultural budgeting uses the same foundational facts—such as farm size or crop type—but applies them through financial models and cost projections, which require a different decision-making framework.

An **ontology** is a formal representation of key entities and their relationships within a domain. For example, in the agriculture domain, entities like crops, soil, and weather conditions are defined, along with relationships such as "crop is grown in a region" or "soil has moisture level." By defining these entities and relationships, the ontology provides a consistent and clear framework for organizing domain knowledge [Guarino et al., 2009, Jackson, 1990]. It differs from *taxonomy* or *classifications* as it allows for richer relationships between entities that need not be hierarchical. More formally,

<样式 id='1'>定义1（本体论）</样式>一个<样式 id='3'>本体论</样式> $\mathcal{O} \subseteq \mathcal{S} \times \mathcal{A} \times$ <样式 id='6'>(</样式> $\mathcal{S} \cup \{\phi\}$ <样式 id='9'>)</样式>由一组三元组组成，这些三元组使用一组属性 $\mathcal{A}$ 将一组实体 $\mathcal{S}$ 关联起来，其中<样式 id='13'>(</样式> s, a, v <样式 id='18'>)</样式> $v \in \mathcal{O}$ 表示主体实体 s 具有属性 a ，而值 v <样式 id='24'>:</样式> $= v_{\mathcal{O}}$ <样式 id='27'>(</样式> s, a <样式 id='31'>)</样式>要么是：

- 另一个实体 $s' \in \mathcal{S}$ ，或
- 一个未指定的领域值，用 ϕ 表示。

在此处， $v := v_{\mathcal{O}}(s, a)$ 表示属性 a 的值，该属性属于实体 s ，该实体是本体中的另一个实体，或未定义（未指定）的文本或数据。

例如，考虑一个主题实体 $s = \text{“作物”}$ ，该实体可以有属性 $a_1 = \text{“生长在”}$ ，将其映射到另一个客体实体 $v_{\mathcal{O}}(s, a_1) = s' = \text{“作物区域”}$ 。此外，相同的实体 s 还可以有另一个属性 $a_2 = \text{“具有名称”}$ ，将其映射到任意文本，表示为 $v_{\mathcal{O}}(s, a_2) = \phi$ ，表明此值未指定，可以是领域中的任何相关文本或名称。

从特定领域的文档中提取事实信息 $\mathcal{D}$ 具有挑战性，因为它们的语言专业且结构通常未充分指定。此外，相关事实经常分散在各个不相关的文档中。本体 $\mathcal{O}$ 提供了一种结构化方法来组织领域内的关键术语及其关系。

为解决此问题，我们提出利用本体论中定义的显式关系从这些文档中提取事实信息。我们主张，由于特定领域的事实通常根植于底层本体论，强制执行这些关系可以帮助丰富和消除文档中包含的信息歧义。特别是，我们使用文档来为属性找到值，方法是提取来自文档本身的与特定领域相关的文本或值（即，当 $v_{\mathcal{O}}(s, a) = \phi$ ）。由于特定领域的文档可能包含各种事实，此值分配不必在所有文档中都是唯一的。相反，文档的不同部分可以提供与同一本体实体相关联的不同但有效的文本/数据值。

因此，我们使用一组自包含的事实块 $F \in \mathcal{D}(\mathcal{O})$ 对提取的信息 $\mathcal{I} := \mathcal{D}(\mathcal{O})$ 进行建模，其中每个事实块 F 包含将本体实体映射到未指定域文本 $\in \mathcal{V}$ 或同一事实块内的另一个实体 F 的关系。更正式地说，

定义 2（本体映射数据） 本体映射数据 $\mathcal{I} := \mathcal{D}(\mathcal{O})$ 是使用本体论 $\mathcal{O}$ 从文档 $\mathcal{D}$ 中提取的信息。它由一组事实块组成，其中每个事实块 F 表示一组本体关系。对于任何关系 $(s, a, v) \in F$ ，其值 v 按如下方式计算：如果值 $v_{\mathcal{O}}(s, a) = \phi$ ，则从文档文本中提取 $v \in \mathcal{V}$ ；否则 $v = v_{\mathcal{O}}(s, a)$ 使用本体提供的值。

因此，本体映射数据表示从特定领域文档中提取的、自包含且基于本体的信息。例如，一个事实块 F 可能表示：术语 $s = \text{“种子”}$ 是 $a_1 = \text{“作物”}$ $v(s, a_1) = \text{“大豆”}$ 是 $a_2 = \text{“生长在”}$ $v(s, a_2) = (s' = \text{“作物区域”})$ ，其 $a_3 = \text{“名称”}$ 为 $v(s', a_3) = \text{“西北地区”}$)。

我们可以应用各种模式匹配启发式算法、基于规则的策略或嵌入相似性来生成这种本体映射数据 [Otero-Cerdeira 等人, 2015, Jackson, 1990]。然而，凭借大语言模型的强大功能，我们利用其自然语言理解能力更有效地将这些本体实体映射到相应的文档文本中 [Babaei Giglou 等人, 2023]。我们通过提示大语言模型以 JSON-LD 格式生成映射信息来完成这项任务。完整的提示内容提供在附录 B.1 中。这种方法的一个局限性是领域特定本体可能并非总是可用或足够全面。为了解决这个问题，我们正在开发一种本体学习方法，可以自动生成一个稳健的基线本体。这为领域专家提供了一个起点，使他们更容易编辑和改进现有本体，而不是从头开始构建。此外，在许多领域，由于数十年来在数据建模和本体开发方面的研究，已经存在丰富的预现有本体，这些本体可以直接被此方法利用。这项工作的细节超出了当前论文的范围，因此我们省略了它们。

4.1.2 超图转换

由于事实块 $F \in \mathcal{D}(\mathcal{O})$ 中定义的嵌套结构，直接处理这些块中的信息具有挑战性。多层关系和依赖的组合性质使得高效提取或归因信息变得困难，这阻碍了我们提供紧凑且准确上下文归因的目标。为此，我们扁平化结构，将本体映射数据 $\mathcal{I}$ 中的每个事实块 F 转换为一组扁平化事实块 F ，从而在不显著丢失细节的情况下简化信息处理。算法1概述了扁平化过程，该过程也如图2所示。

Definition 1 (Ontology) An *ontology* $\mathcal{O} \subseteq \mathcal{S} \times \mathcal{A} \times (\mathcal{S} \cup \{\phi\})$ consists of a set of triples that relate a set of entities $\mathcal{S}$ using a set of attributes $\mathcal{A}$, where $(s, a, v) \in \mathcal{O}$ denotes that the subject entity s has an attribute a , and the value $v := v_{\mathcal{O}}(s, a)$ is either:

- Another entity $s' \in \mathcal{S}$, or
- An unspecified domain value, denoted by ϕ .

Here, $v := v_{\mathcal{O}}(s, a)$ represents the value of the attribute a for entity s , which is either another entity within the ontology or an undefined (unspecified) text or data.

For example, consider a subject entity $s = \text{“Crop”}$, that can have the attribute $a_1 = \text{“is grown in”}$, which maps it to another object entity $v_{\mathcal{O}}(s, a_1) = s' = \text{“Crop Region”}$. Additionally, the same entity s can have another attribute $a_2 = \text{“has name”}$, which maps it to an arbitrary text, denoted as $v_{\mathcal{O}}(s, a_2) = \phi$, indicating that this value is unspecified and can be any relevant text or name in the domain.

Extracting factual information from domain-specific documents $\mathcal{D}$ is challenging due to their specialized language and often underspecified structure. Moreover, relevant facts are frequently scattered across various unrelated documents. An ontology $\mathcal{O}$ provides a structured way to organize key terms and their relationships within a domain.

To address this, we propose leveraging the explicit relationships defined in the ontology to extract factual information from these documents. We claim that since domain-specific facts are often grounded in the underlying ontology, enforcing these relationships can help enrich and disambiguate the information contained in the documents. In particular, we use the documents to find values for attributes by extracting relevant domain-specific text or values from the documents themselves (*i.e.*, when $v_{\mathcal{O}}(s, a) = \phi$). Since domain-specific documents may contain a variety of facts, this value assignment does not have to be unique across all documents. Instead, different parts of the documents may provide distinct yet valid text/data value related to the same ontology entity.

Therefore, we model the extracted information $\mathcal{I} := \mathcal{D}(\mathcal{O})$ using a set of self-contained *factual-blocks* $F \in \mathcal{D}(\mathcal{O})$, where each *factual-block* F consists of relationships that map ontology entities to either a unspecified domain text $\in \mathcal{V}$ or another entity within the same factual-block F . More formally,

Definition 2 (Ontology-mapped data) *Ontology-mapped data* $\mathcal{I} := \mathcal{D}(\mathcal{O})$ is information derived from the documents $\mathcal{D}$ using the ontology $\mathcal{O}$. It consists of a set of *factual-blocks*, where each *factual-block* F represents a set of ontology relationships. For any relationship $(s, a, v) \in F$, the value v is derived as follows: If value $v_{\mathcal{O}}(s, a) = \phi$ then $v \in \mathcal{V}$ is extracted from the document text; otherwise $v = v_{\mathcal{O}}(s, a)$ is the value provided by the ontology.

Thus, ontology-mapped data represents *self-contained* and *ontology-grounded* information extracted from domain-specific documents. For example, a factual-block F might represent that: a term $s = \text{“Seed”}$ is $a_1 = \text{“of crop”}$ $v(s, a_1) = \text{“Soybean”}$ is $a_2 = \text{“is grown in”}$ $v(s, a_2) = (s' = \text{“Crop Region”})$, which $a_3 = \text{“has a name”}$ of $v(s', a_3) = \text{“Northwest Region”}$).

We can apply various pattern-matching heuristics, rule-based strategies, or embedding similarity to generate this ontology-mapped data [Otero-Cerdeira et al., 2015, Jackson, 1990]. However, with the powerful capabilities of LLMs, we leverage their natural language understanding capabilities to map these ontology entities to corresponding document text more effectively [Babaei Giglou et al., 2023]. We perform this task by prompting the LLM to generate the mapped information in JSON-LD format. The complete prompt is provided in Appendix B.1.

One limitation of this method is that domain-specific ontologies may not always be available or sufficiently comprehensive. To address this, we are developing an *ontology learning* method that can automatically generate a robust baseline ontology. This provides domain experts with a starting point, making it easier for them to edit and refine an existing ontology rather than building one from scratch. Additionally, in many fields, rich pre-existing ontologies are already available due to decades of research in data modeling and ontology development, which can be directly leveraged by this method. The details of this work are beyond the scope of the current paper, so we omit them.

4.1.2 Hypergraph Transformation

Due to the nested structure of the definitions in the factual-block $F \in \mathcal{D}(\mathcal{O})$, directly processing the information in these blocks is challenging. The combinatorial nature of the multi-layered relationships and dependencies make it difficult to efficiently extract or attribute information, which hampers our goal of providing compact and accurate context attribution. To address this, we flatten the structure so that each factual-block F in the ontology-mapped data $\mathcal{I}$ is converted into a set of flattened factual-blocks $\overline{F}$, making the information easier to handle without significant loss of detail. Algorithm 1 outlines the flattening process, which is also illustrated in Figure 2.

算法 1 扁平化事实块

需求: 事实块 F , 连接运算符 $\oplus$ 。

保证: 扁平化事实块集合 $F \leftarrow \text{FLATTEN}(F)$ 扁平化 F 中的任何嵌套信息。

```

1: 过程 FLATTEN( $F$ )
2:    $\overline{F} \leftarrow \{\}$ 
3:    $\overline{F}_0 \leftarrow \{(s \oplus a, v) : (s, a, v) \in F, v \in \mathcal{V}, (s', a', s) \notin F\}$ .      ▷ 没有依赖, 可以直接扁平化
4:    $\overline{F} \leftarrow \overline{F} \cup \{\overline{F}_0\}$ 
5:   对于  $(s, a, s') \in F \setminus \overline{F}_0$  做。
6:     如果  $s' \in \mathcal{S}$  则
7:        $F_{s'} \leftarrow \overline{F}_0 \cup \{(s \oplus a \oplus s' \oplus a', v') : (s', a', v') \in F\}$ .
8:        $\overline{F} \leftarrow \overline{F} \cup \text{FLATTEN}(F_{s'})$ .      ▷ 展平嵌套的  $s'$ 
9:     结束if
10:  结束for
11:  返回  $\overline{F}$ 
12: 结束过程

```

我们将每个扁平化事实块定义为一个超边 $e \in \mathcal{E}$, 其中超边 e 连接多个超节点 $\{n_i \in \mathcal{N}\}$, 每个超节点 $n_i \in F$ 是扁平化块中的一个原始集合, 可以表示为键值对。重要的是, 此扁平化过程保持了信息的完整性, 而未引入数据丢失。这使得 OG-RAG 能够捕获事实之间的多维关系, 而不同于仅处理成对连接的简单图模型。我们现在可以将从本体中提取的信息转换为更结构化的超图, 其定义如下:

定义3 (超图) 一个超图 $\mathcal{H} := (\mathcal{N}, \mathcal{E})$ 由超节点 $\mathcal{N}$ 和超边 $\mathcal{E}$ 组成, 其中每个超边 $e \in \mathcal{E}$ 是一组任意长度的节点集合。定义 $\mathcal{P}(X)$ 为 X 的幂集, 而 $\bigoplus X$ 是由连接集合 X 中每个元素内的字符串形成的集合, 因此我们得到超边 $\mathcal{E} \subseteq \mathcal{P}(\mathcal{N})$ 和超节点 $\mathcal{N} \subseteq [\bigoplus \mathcal{P}(\mathcal{S} \times \mathcal{A})] \times \mathcal{V}$, 其中 $\times$ 是笛卡尔积。

因此, 从本体映射文档中提取的所有扁平化事实块 $\mathcal{I}$ 可以被视为一个超图。我们将其简单地称为 $\mathcal{H}(\mathcal{I})$ 。根据此定义, 超节点本质上是一个键值对, 我们声明超边是建立在领域特定数据上的真实事实。从数学上讲,

定义 4 (事实) 事实是在两个实体——主体和客体之间, 通过一个功能属性所形成的逻辑断言, 该断言可以被证据验证为真或假。形式上, 它可以表示为一个可以被验证为具有真或假值的逻辑断言。例如, 考虑断言: $\text{hasCropYield}(\text{Farm } A) = 500\text{吨}$, 其中 hasCropYield 是映射农场 (主体) 到作物产量 (值) 的功能属性, 并且可以被证据验证为真或假。

因此, 在 OG-RAG 中, 超边可以被视为一个复杂事实的表示。不失一般性, 考虑两个超节点, $n_1(s_1 \oplus a_1, v_1) = (\text{作物名称, 大豆})$ 和 $n_2(p_2 \in \bigoplus \mathcal{P}(\mathcal{S} \times \mathcal{A}) v_2) = (\text{作物生长区作物种植区名称, 西北})$ 形成超边 $e = ((\text{作物名称, 大豆}), (\text{作物生长区作物种植区名称, 西北}))$ 可以表示为一个简化事实:

$$\text{hasGrowingZone}(\text{Crop has name Soybean}) = \text{Northwest},$$

可以被证据验证为真或假。

这样, OG-RAG 超图构建能够对适用于特定领域的事实进行紧凑准确的表示。这种结构通过允许用户检查封装了实体之间关系和依赖的超边, 从而促进事实验证。

4.2 基于超图的检索

在基于领域特定信息构建的超图上, 即, $\mathcal{H}(\mathcal{I}(\mathcal{D}, \mathcal{O}))$, OG-RAG 现在可以根据用户查询 Q 来检索相关上下文, 以支持大语言模型生成准确、特定领域的响应。

Algorithm 1 Flattening a factual block

Require: Factual block F , Concatenation operator $\oplus$.

Ensure: A set of flattened factual-blocks $\overline{F} \leftarrow \text{FLATTEN}(F)$ flattens any nested information present in F .

```

1: procedure FLATTEN( $F$ )
2:    $\overline{F} \leftarrow \{\}$ 
3:    $\overline{F}_0 \leftarrow \{(s \oplus a, v) : (s, a, v) \in F, v \in \mathcal{V}, (s', a', s) \notin F\}$ .      ▷ no dependencies, can be directly flattened
4:    $\overline{F} \leftarrow \overline{F} \cup \{\overline{F}_0\}$ .
5:   for  $(s, a, s') \in F \setminus \overline{F}_0$  do.
6:     if  $s' \in \mathcal{S}$  then
7:        $F_{s'} \leftarrow \overline{F}_0 \cup \{(s \oplus a \oplus s' \oplus a', v') : (s', a', v') \in F\}$ .
8:        $\overline{F} \leftarrow \overline{F} \cup \text{FLATTEN}(F_{s'})$ .      ▷ flatten nesting of  $s'$ 
9:     end if
10:  end for
11:  return  $\overline{F}$ 
12: end procedure

```

We define each flattened factual-block as a hyperedge $e \in \mathcal{E}$, where a hyperedge e connects multiple hypernodes $\{n_i \in \mathcal{N}\}$, where each hypernode $n_i \in \overline{F}$ is a primitive set in the flattened-block that can be represented as a key-value pair. Importantly, this flattening process maintains the integrity of the information without introducing data loss. This allows OG-RAG to capture multi-dimensional relationships between facts, unlike simpler graph-based models that only handle pairwise connections. We can now convert the extracted information from the ontology into a more structured hypergraph, defined as follows:

Definition 3 (Hypergraph) A hypergraph $\mathcal{H} := (\mathcal{N}, \mathcal{E})$ consists of hypernodes $\mathcal{N}$ and hyperedges $\mathcal{E}$, such that each hyperedge $e \in \mathcal{E}$ is a set of nodes with arbitrary length. Defining $\mathcal{P}(X)$ as the power set of X and $\bigoplus X$ as the set that is formed by concatenating the strings within each element of the set X , we have the hyperedges $\mathcal{E} \subseteq \mathcal{P}(\mathcal{N})$ and the hypernodes $\mathcal{N} \subseteq [\bigoplus \mathcal{P}(\mathcal{S} \times \mathcal{A})] \times \mathcal{V}$, where $\times$ is the cartesian product.

Thus, the set of all flattened factual-blocks extracted from the ontology-mapped documents $\overline{\mathcal{I}}$ thus can be seen as a hypergraph. We call this simply $\mathcal{H}(\mathcal{I})$. With this definition, a hypernode is essentially a key-value pair and we declare a hyperedge to be a true *fact* grounded in domain-specific data. Mathematically,

Definition 4 (Fact) A fact is a logical assertion between two entities - subject and object, through a functional attribute, which can be evidentially verified to be either true or false. Formally, it can be expressed as a logical assertion that can be verified to have a value of True or False. For example, consider the assertion: $\text{hasCropYield}(\text{Farm } A) = 500\text{ tons}$, where hasCropYield is the functional attribute mapping a farm (subject) to a crop yield (value), and which can be evidentially verified to be either True or False.

Therefore, in OG-RAG a hyperedge can be viewed as a representation of a complex fact. Without loss of generality, consider two hypernodes, $n_1(s_1 \oplus a_1, v_1) = (\text{Crop has name, Soybean})$ and $n_2(p_2 \in \bigoplus \mathcal{P}(\mathcal{S} \times \mathcal{A}), v_2) = (\text{Crop has growing zone CropGrowingZone with name, Northwest})$ forming an hyperedge $e = ((\text{Crop has name, Soybean}), (\text{Crop has growing zone CropGrowingZone with name, Northwest}))$ can be represented as a simplified fact:

$$\text{hasGrowingZone}(\text{Crop has name Soybean}) = \text{Northwest},$$

which can be evidentially verified to be True or False.

In this way, the OG-RAG hypergraph construction enables a compact and accurate representation of *facts* that are adapted to the specific domain. This structure facilitates fact verification by allowing users to inspect the hyperedges, which encapsulate the relationships and dependencies between entities.

4.2 Hypergraph-based retrieval

With the hypergraph constructed on domain-specific information, *i.e.*, $\mathcal{H}(\mathcal{I}(\mathcal{D}, \mathcal{O}))$, OG-RAG is now ready to retrieve relevant context based on user query Q that can support the LLM in generating accurate, domain-specific responses.

算法2 本体引导检索增强生成

需求: 查询 Q 、领域特定本体 $\mathcal{O}$ 、文档 $\mathcal{D}$ 、句子嵌入函数 $\mathbf{Z}$ 、大语言模型 $\mathcal{M}_0$ 、最大长度 L

保证: 检索上下文 $\mathcal{C}_{\mathcal{H}}(Q)$ 在本体中具有基础且与查询相关

```

1: 过程 OG-PREPROCESS( $\mathcal{O}, \mathcal{D}, \mathcal{M}_0$ )
2:    $\mathcal{I} \leftarrow$  大语言模型  $\mathcal{M}_0$ (本体论映射  $(\mathcal{D}, \mathcal{O})$ ) ▷ 参见定义2
3:    $\mathcal{H}(\mathcal{I}) \leftarrow$  超图带边  $\bigcup_{F \in \mathcal{I}} \text{FLATTEN}(F)$ .
4: 结束流程
5: 流程 OG-检索( $Q, \mathcal{H}(\mathcal{I}), \mathbf{Z}, k, L$ )
6:    $\mathcal{N}, \mathcal{E} \leftarrow$  超图的节点和边  $\mathcal{H}(\mathcal{I})$ .
7:    $\mathcal{N}_S(Q) \leftarrow$  top-k arg max $_{(s,a,v) \in \mathcal{N}} \langle \mathbf{Z}(s \oplus a), \mathbf{Z}(Q) \rangle$ .
8:    $\mathcal{N}_V(Q) \leftarrow$  top-k arg max $_{(s,a,v) \in \mathcal{N}} \langle \mathbf{Z}(v), \mathbf{Z}(Q) \rangle$ 
9:    $\mathcal{N}(Q) \leftarrow \mathcal{N}_S(Q) \cup \mathcal{N}_V(Q)$ .
10:   $\mathcal{C}_{\mathcal{H}}(Q) \leftarrow \{\}$ 
11:  当 ( $|\mathcal{N}(Q)| > 0$ )  $\vee$  ( $|\mathcal{C}_{\mathcal{H}}(Q)| < L$ ) 时
12:     $\mathcal{C}_{\mathcal{H}}(Q) \leftarrow \mathcal{C}_{\mathcal{H}}(Q) \cup \text{最大值}_{e \in \mathcal{E}} |\{n \in \mathcal{N}(Q) : n \in e\}|$ 
13:  结束 while
14:  返回  $\mathcal{C}_{\mathcal{H}}(Q)$ 
15: 结束过程
```

4.2.1 相关节点

我们首先识别与给定查询相关的超节点集。使用定义 3，超节点 $n \in \mathcal{N}$ 可以表示为一个来自集合 $\mathcal{S}$ 、 $\mathcal{A}$ 、 $\mathcal{V}$ 中元素的键值对。然后，如果超节点与查询相关，则满足：(1) 查询关联于术语 s 的属性 a ，或 (2) 查询关注具有特定值 v 的对象。换句话说，如果键（表示连接的实体和属性）与查询 Q 之间的相似性较高，或者值 v 与查询 Q 之间的相似性较高，则超节点是相关的。OG-RAG 找到两组与查询相关的超节点： $\mathcal{N}_S(Q)$ 和 $\mathcal{N}_V(Q)$ ，分别表示这两个集合。特别是， $\mathcal{N}_S(Q)$ 表示在向量空间 $\mathbf{Z}$ 中，其属性术语（即 $s \oplus a$ ）与查询 Q 具有最高相似性的前 k 个超节点。类似地， $\mathcal{N}_V(Q)$ 表示其值 v 与查询 Q 具有最高相似性的前 k 个超节点。因此，对于每个查询，我们提取 $2 \cdot k$ 个相关超节点。

4.2.2 相关超边作为上下文

我们将相关上下文定义为最小覆盖相关超节点的超边集合 $\mathcal{C}_{\mathcal{H}}(Q) \subset \mathcal{E}$ ，即 $\mathcal{N}(Q) = \mathcal{N}_S(Q) \cup \mathcal{N}_V(Q)$ 。这被表述为一个优化问题，并以贪婪方式求解。由于在格约束下最小化超边数量的目标函数是线性的，因此可以证明该解是最优的 [Korte et al., 2011]。具体来说，我们维护一个字典，将每个超节点 $n \in \mathcal{N}$ 映射到它所属的超边集合，即 $\mathcal{E}(n)$ ，其中 $e \in \mathcal{E}(n) \implies n \in e$ 。在每次迭代中，我们将覆盖最多未覆盖节点的超边添加到上下文中，并从进一步考虑中移除这些节点。重复此过程，直到达到 L 个超边或所有相关节点都被覆盖。通过这种方式，上下文被构建为最多包含 L 个超边的集合，这些超边代表与给定查询相关的事实。通过将信息组织成超边，OG-RAG 能够将相关事实分组在一起，确保检索到的上下文既紧凑又全面，捕捉所有必要的事实以支持准确的 LLM 响应，同时优化效率。

4.2.3 检索增强生成

最后，给定一个用户查询 Q 以及在上面找到的相关上下文，我们提示大语言模型 $\mathcal{M}$ 使用这个上下文来回答查询，作为 $\mathcal{M}(\mathcal{P}(Q), \mathcal{C}_{\mathcal{H}}(Q))$ ，其中 $\mathcal{P}$ 表示相应的文本提示：

给定下面的上下文，生成给定查询的答案。请注意，上下文以字典格式的有效事实列表提供。

上下文: $\langle$ 按行分隔的检索上下文 $\mathcal{C}_{\mathcal{H}}(Q)$ $\rangle$ 查询: $\langle$ 用户定义的查询 Q $\rangle$ 答案:

Algorithm 2 Ontology-grounded Retrieval Augmented Generation

Require: Query Q , Domain-specific Ontology $\mathcal{O}$, Documents $\mathcal{D}$, Sentence embedding function $\mathbf{Z}$, LLM $\mathcal{M}_0$, Maximum length L

Ensure: Retrieved context $\mathcal{C}_{\mathcal{H}}(Q)$ is grounded in the ontology and relevant to the query

```

1: procedure OG-PREPROCESS( $\mathcal{O}, \mathcal{D}, \mathcal{M}_0$ )
2:    $\mathcal{I} \leftarrow$  LLM  $\mathcal{M}_0$ (Ontology Map  $(\mathcal{D}, \mathcal{O})$ ) ▷ See Definition 2
3:    $\mathcal{H}(\mathcal{I}) \leftarrow$  Hypergraph with edges  $\bigcup_{F \in \mathcal{I}} \text{FLATTEN}(F)$ .
4: end procedure
5: procedure OG-RETRIEVE( $Q, \mathcal{H}(\mathcal{I}), \mathbf{Z}, k, L$ )
6:    $\mathcal{N}, \mathcal{E} \leftarrow$  nodes and edges of the hypergraph  $\mathcal{H}(\mathcal{I})$ .
7:    $\mathcal{N}_S(Q) \leftarrow$  top-k arg max $_{(s,a,v) \in \mathcal{N}} \langle \mathbf{Z}(s \oplus a), \mathbf{Z}(Q) \rangle$ .
8:    $\mathcal{N}_V(Q) \leftarrow$  top-k arg max $_{(s,a,v) \in \mathcal{N}} \langle \mathbf{Z}(v), \mathbf{Z}(Q) \rangle$ .
9:    $\mathcal{N}(Q) \leftarrow \mathcal{N}_S(Q) \cup \mathcal{N}_V(Q)$ .
10:   $\mathcal{C}_{\mathcal{H}}(Q) \leftarrow \{\}$ 
11:  while ( $|\mathcal{N}(Q)| > 0$ )  $\vee$  ( $|\mathcal{C}_{\mathcal{H}}(Q)| < L$ ) do
12:     $\mathcal{C}_{\mathcal{H}}(Q) \leftarrow \mathcal{C}_{\mathcal{H}}(Q) \cup \arg \max_{e \in \mathcal{E}} |\{n \in \mathcal{N}(Q) : n \in e\}|$ 
13:  end while
14:  return  $\mathcal{C}_{\mathcal{H}}(Q)$ 
15: end procedure
```

4.2.1 Relevant Nodes

We first identify the set of hypernodes relevant to a given query. Using Definition 3, a hypernode $n \in \mathcal{N}$ can be represented as a key-value pair that comes from the elements in the sets $\mathcal{S}, \mathcal{A}, \mathcal{V}$. A hypernode can then be considered relevant to a query if: (1) the query pertains to an attribute a of the term s , or (2) the query focuses on an object with specific values v . In other words, a hypernode is relevant if either the similarity between the key (representing concatenated entities and attributes) and the query Q is high, or the similarity between v (the value) and the query Q is high. OG-RAG finds two sets of query-relevant hypernodes: $\mathcal{N}_S(Q)$ and $\mathcal{N}_V(Q)$ to represent the two sets respectively. In particular, $\mathcal{N}_S(Q)$ denotes the top k hypernodes with the highest similarity between their attributed term, i.e., $s \oplus a$ and the query Q in the vector space $\mathbf{Z}$. Similarly, $\mathcal{N}_V(Q)$ represents the top k hypernodes with the highest similarity between their value v and the query Q . Thus, for each query, we extract $2 \cdot k$ relevant hypernodes.

4.2.2 Relevant Hyeredges as Context

We form the relevant context as the set of hyperedges $\mathcal{C}_{\mathcal{H}}(Q) \subset \mathcal{E}$ that minimally cover the relevant hypernodes, $\mathcal{N}(Q) = \mathcal{N}_S(Q) \cup \mathcal{N}_V(Q)$. This is formulated as an optimization problem and solved in a greedy manner. Since the objective of minimizing the number of hyperedges is linear under a matroid constraint, the solution can be shown to be optimal [Korte et al., 2011]. Specifically, we maintain a dictionary that maps each hypernode $n \in \mathcal{N}$ to the set of hyperedges that it is a part of, i.e., $\mathcal{E}(n)$, where $e \in \mathcal{E}(n) \implies n \in e$. In each iteration, we add the hyperedge that covers the largest number of uncovered nodes to the context and remove those nodes from further consideration. This process is repeated until either we have L hyperedges or all the relevant nodes are covered. In this way, the context is constructed as a collection of up to L hyperedges representing *facts* relevant to the given query. By organizing information into hyperedges, OG-RAG is able to group related facts together, ensuring that the retrieved context is both compact and comprehensive, capturing all necessary facts to support accurate LLM responses, while optimizing for efficiency.

4.2.3 Retrieval-Augmented Generation

Finally, given a user query Q and the relevant context as found above, we prompt the LLM $\mathcal{M}$ to use this context to answer the query as $\mathcal{M}(\mathcal{P}(Q), \mathcal{C}_{\mathcal{H}}(Q))$, where $\mathcal{P}$ denotes the corresponding textual prompt:

Given the context below, generate the answer to the given query. Note that the context is provided as a list of valid facts in a dictionary format.
Context: $\langle$ Line-separated retrieved context $\mathcal{C}_{\mathcal{H}}(Q)$ $\rangle$
Query: $\langle$ User-defined query Q $\rangle$
Answer:

4.3 复杂度分析

算法2概述了所提出方法的完整流程，该方法由两个主要组件组成：(1) OG-预处理，对文档集应用一次，以及 (2) OG-检索，用于检索每个查询的相关上下文。

4.3.1 查询复杂度

假设大语言模型 $\mathcal{M}_0$ 的上下文大小为 N_C 。本体论 $\mathcal{O}$ 可以用 JSON-LD 或文本格式表示，其长度为 $|\mathcal{O}|$ ，其中属性映射到自然语言词汇表中的相应范围。OG-PREPROCESS 阶段可能涉及多个大语言模型调用，具体取决于文档块的数量，即 $(|\mathcal{D}| + |\mathcal{O}|)/N_C$ 次调用。在 OG-RETRIEVE 程序的查询时间内，我们不进行任何额外的调用。

4.3.2 时间复杂度

在计算时间复杂度时，我们忽略了大语言模型调用的耗时，因为这部分已经包含在查询复杂度中。因此，OG-PREPROCESS 步骤的时间复杂度仅涉及通过展平映射数据来转换超图。假设我们从文档中获得了 $|\mathcal{I}|$ 个事实块，并且每个事实块的最大长度为 $|F|_{max} = O(|\mathcal{O}|)$ 。我们考虑两种情况：(1) **最小或无嵌套**：在这种情况下，时间复杂度由算法中的第 4 步决定，导致复杂度为 $O(|\mathcal{O}||\mathcal{I}|)$ ，(2) **最大嵌套**：在这种情况下，第 4 步可能产生空集。因此，每个事实块 F 在遍历整个集合时可以递归地展平 $\log |\mathcal{O}|$ 次，导致时间复杂度为 $O(|\mathcal{I}||\mathcal{O}|\log |\mathcal{O}|)$ 。

4.3.3 空间复杂度

所需的唯一存储空间是用于超图结构 $\mathcal{H}(\mathcal{I})$ ，它直接与超边数量 $|\mathcal{E}| = |\mathcal{I}|$ 成正比。

5 实验设置

数据集。 我们在涉及专业工作流的两个不同领域类别中评估了OG-RAG：(a) 工业工作流，重点关注农业领域，其中精确的、数据驱动的决策对于作物管理和资源分配至关重要，以及(b) 知识工作，我们在新闻领域的科研和分析任务上评估它。我们避免使用维基百科等通用领域，以减轻潜在的数据污染对大语言模型训练的影响。对于农业领域，我们使用了两个由农业专家准备的包含85份文档的专有高质量数据集，重点关注印度的大豆和小麦作物栽培。对于新闻领域，我们使用了来自Multi-hop RAG [Tang和Yang, 2024], 的公开数据集，并筛选出149篇长篇文章（每篇超过2,000字），这些文章聚焦于需要详细、丰富上下文分析的、多方面、复杂的新闻故事。请参考附录A获取数据集的示例摘录。

本体论。 我们采用半自动化方法构建两个领域的本体，这反映了OG-RAG在专业工作流程中的更广泛适用性。对于农业领域，本本体是使用专有的本体学习模块生成的，然后由多位专门从事作物栽培的专家进行审查和验证。对于新闻领域，我们修改了现有的简单新闻与新闻本体 (SNaP)¹。具体来说，我们通过排除某些类（例如与图像、视频以及“内容”层次结构相关的类）来简化其结构。相反，我们允许一个资产链接到多个事件，并且每个事件可以与多个组织和人物相关联。两个领域的完整本体都在附录B.2中提供。

大语言模型。 我们考虑 4 个大语言模型用于零样本查询回答，同时添加从不同方法检索到的上下文：2 个封闭模型² (GPT-4o-mini 和 GPT-4o) 和 2 个开源模型³ (Llama-3.1-8B 和 Llama-3.1-70B)。这些模型因其卓越的理解能力和自然语言推理能力而被选择。我们考虑 4096 个完成 token 和温度为 0。

基线方法。 我们比较将OG-RAG与三种领先的检索方法进行对比，这些方法代表了上下文检索领域的最先进技术针对上下文检索生成，以展示其有效性。

¹<https://iptc.org/thirdparty/snap-ontology/>
²<https://openai.com/index>
³<https://ai.meta.com/blog/meta-llama-3-1/>

4.3 Complexity Analysis

Algorithm 2 outlines the full procedure of the proposed method which consists of two main components: (1) OG-PREPROCESS, applied to the set of documents once, and (2) OG-RETRIEVE, used to retrieve the relevant context for each query.

4.3.1 Query Complexity

Assume the context size for the LLM $\mathcal{M}_0$ is N_C . The ontology $\mathcal{O}$, which can be written in a JSON-LD or textual format, has a length $|\mathcal{O}|$, where the attributes are mapped to their corresponding ranges in the natural language vocabulary. OG-PREPROCESS phase may involves several LLM calls depending on the number of document chunks, specifically, $(|\mathcal{D}| + |\mathcal{O}|)/N_C$ number of calls. We do not make any additional LLM calls during the querying time in the OG-RETRIEVE procedure.

4.3.2 Time Complexity

We ignore the time taken by LLM calls while calculating the time complexity, as this is accounted for under query complexity. Thus, the time complexity of the OG-PREPROCESS step only involves the hypergraph transformation by flattening the mapped data. Let us assume we have $|\mathcal{I}|$ factual-blocks derived from the documents, and each factual-block has a maximum length of $|F|_{max} = O(|\mathcal{O}|)$. We consider two cases: (1) **Minimal or No Nesting**: In this case, the time complexity is determined by step 4 in the algorithm, leading to a complexity of $O(|\mathcal{O}||\mathcal{I}|)$, (2) **Maximum Nesting**: In this scenario, step 4 may result in an empty set. Thus, each factual-block F can be recursively flattened $\log |\mathcal{O}|$ times while searching through the entire set, leading to a time complexity of $O(|\mathcal{I}||\mathcal{O}|\log |\mathcal{O}|)$.

4.3.3 Space Complexity

The only storage required is for the hypergraph structure $\mathcal{H}(\mathcal{I})$, which is directly proportional to the number of hyperedges $|\mathcal{E}| = |\mathcal{I}|$.

5 Experimental Setup

Datasets. We evaluate OG-RAG across two distinct domain categories that involve specialized workflows: (a) Industrial workflows, with a focus on the agriculture domain, where precise, data-driven decisions are critical for crop management and resource allocation, and (b) Knowledge work, where we evaluate it on research and analysis tasks in the news domain. We avoid general domains like Wikipedia to mitigate potential data contamination in LLM training. For the agriculture domain, we utilize two proprietary high-quality datasets comprising of 85 documents prepared by agriculture experts, focusing on the crop cultivation of Soybean and Wheat in India. For the news domain, we use the publicly available dataset from Multi-hop RAG [Tang and Yang, 2024], filtered for 149 long-form articles (each over 2,000 words) focused on multi-faceted, complex news stories requiring detailed, contextually rich analysis. Please refer to Appendix A for exemplary excerpts from the datasets.

Ontology. We use a semi-automated approach to construct the ontology for both domains, which reflects the broader applicability of OG-RAG in specialized workflows. For the agriculture domain, the ontology was generated using a proprietary ontology learning module, which was then reviewed and verified by multiple experts specializing in crop cultivation. For the news domain, we modify the existing Simple News and Press (SNaP) ontology¹. Specifically, we simplify its structure by excluding certain classes, such as those related to images, videos, and the "stuff" hierarchy. Instead, we allow an asset to be linked to multiple events, and each event can be associated with multiple organizations and persons. The complete ontologies for both domains are provided in Appendix B.2.

Large Language Models. We consider 4 large language models for zero-shot query answering while adding the retrieved context from different methods: 2 closed-box models² (GPT-4o-mini and GPT-4o) and 2 open-source models³ (Llama-3.1-8B and Llama-3.1-70B). These models have been chosen for their remarkable understanding and ability to reason in natural language. We consider 4096 completion tokens and a temperature of 0.

Baselines. We compare OG-RAG against three leading retrieval-based methods, representing state-of-the-art approaches to context retrieval and generation, to demonstrate its effectiveness.

¹<https://iptc.org/thirdparty/snap-ontology/>
²<https://openai.com/index>
³<https://ai.meta.com/blog/meta-llama-3-1/>

表1: 不同方法在特定领域问答中检索到的上下文质量。我们发现所有指标95%置信区间为 ≤ 0.05 , 表示误差范围很小。此处未报告。符号 ‘-’ 表示计算在1天内未完成。

方法	大豆					小麦					News				
	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel
RAG	0.22	0.08	0.31	0.62	0.29	0.14	0.04	0.29	0.69	0.28	0.01	0.01	0.27	0.67	0.20
RAPTOR	0.54	0.19	0.34	0.68	0.68	0.85	0.29	0.59	0.79	0.89	0.82	0.46	0.58	0.84	0.76
GraphRAG	0.41	0.14	0.26	0.63	0.63	0.78	0.05	-	-	-	-	-	-	-	-
OG-RAG	0.84	0.41	0.48	0.72	0.79	0.95	0.34	0.62	0.79	0.79	0.82	0.52	0.66	0.86	0.73

1. **RAG [Lewis等人, 2020]:** RAG (检索增强生成) 通过将查询相关文档片段嵌入到向量空间中, 然后根据最大片段-查询相似性找到上下文来检索查询相关文档片段。
2. **RAPTOR [Sarathi等人, 2024]:** RAPTOR将文档片段聚类成层次结构, 并使用大语言模型对聚类进行总结作为附加上下文。对于此实验, 我们将树深度设置为3, 并使用折叠树检索策略。

3. **GraphRAG [Edge等人, 2024]:** GraphRAG 从使用大语言模型构建的知识图谱中检索信息, 通过提取实体和关系并将它们聚类成语义社区。我们使用默认的图谱构建提示, 并以社区级别为2进行本地搜索以进行检索。

我们使用 text-embedding-3-small ² 作为所有检索方法的句子嵌入函数, 并使用 GPT-4o 作为大语言模型 (即, $\mathcal{M}_0$) 进行预处理。对于每种方法, 我们找到 $\{2, 5\}$ 相似的上下文, 并选择性能最高的那个。

指标。基于 RAGAS 框架 [RAGAS, 2024],我们使用以下指标来评估检索上下文和生成响应的质量, 同时使用 text-embedding-3-small 作为嵌入模型, 并使用 GPT-4o 作为大语言模型。

1. **上下文召回 (C-Rec):** 地面真实答案中可以归因于检索上下文中信息的声明比例。
2. **上下文实体召回 (C-ERec):** 地面真实答案中存在于检索上下文中的实体比例。
3. **答案相似度 (A-Sim):** 生成响应与嵌入空间中地面真实答案之间的相似度。
4. **答案正确性 (A-Corr):** 答案相似度 (如上所述) 和事实相似度的组合, 即地面真实答案中的声明与生成响应中声明之间的 F1 分数。
5. **答案相关性 (A-Rel):** 衡量从生成响应中推断出原始问题的容易程度。

伦理考量。该调查符合ACM关于人类研究的出版物政策。未收集任何个人数据, 所有回复均已匿名化。对于农业和新闻领域的实验, 仅使用了公开可用或专有数据集, 未涉及敏感数据, 符合伦理研究标准。

6 实验

6.1 查询回答

6.1.1 问题生成

我们使用 RAGAS 框架 [RAGAS, 2024] 生成一组问答对, 以验证我们提出的方法的事实准确性。RAGAS 提示现成的 LLM 生成不同难度的问题, 每个问题都有相应的真实答案和上下文。具体来说, 我们从 RAGAS 生成了最多 100 个独特的、专注于多跳推理能力的问题, 这在专业领域任务中通常是需要。这些生成问题的示例及其真实答案在附录 C 中提供。

6.1.2 OG-RAG 是否有助于检索有用上下文?

如果上下文为查询提供了足够的信息来推导出真实响应, 则被认为是有用的。我们使用上下文召回率和上下文实体召回率来评估这一点。表1比较了不同检索方法在三个数据集上的性能。

Table 1: Quality of contexts retrieved by different methods for domain-specific query-answering. We found the 95% confidence interval to be ≤ 0.05 for all metrics, representing small margin of error. It is not reported here. The symbol ‘-’ denotes that the computation did not complete within 1 day.

Method	Soybean					Wheat					News				
	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel
RAG	0.22	0.08	0.31	0.62	0.29	0.14	0.04	0.29	0.69	0.28	0.01	0.01	0.27	0.67	0.20
RAPTOR	0.54	0.19	0.34	0.68	0.68	0.85	0.29	0.59	0.79	0.89	0.82	0.46	0.58	0.84	0.76
GraphRAG	0.41	0.14	0.26	0.63	0.63	0.78	0.05	-	-	-	-	-	-	-	-
OG-RAG	0.84	0.41	0.48	0.72	0.79	0.95	0.34	0.62	0.79	0.79	0.82	0.52	0.66	0.86	0.73

1. **RAG [Lewis et al., 2020]:** RAG (Retrieval-Augmented Generation) retrieves query-relevant document chunks by embedding them into a vector space and then finding the context based on the maximum chunk-query similarity.
2. **RAPTOR [Sarathi et al., 2024]:** RAPTOR clusters document chunks into hierarchical structures and uses an LLM to summarize the clusters as additional context. For this experiment, we set the tree depth to 3 and use the collapsed-tree retrieval strategy.
3. **GraphRAG [Edge et al., 2024]:** GraphRAG retrieves from a knowledge graph constructed using an LLM by extracting entities and relationships and clustering them into semantic communities. We use default graph construction prompts and local search with community level as 2 for retrieval.

We use the text-embedding-3-small ² as the sentence embedding function across all retrieval methods and GPT-4o as the LLM (*i.e.*, $\mathcal{M}_0$) for pre-processing. For each method, we find $\{2, 5\}$ similar contexts and select the one with the highest performance.

Metrics. Building on the RAGAS framework [RAGAS, 2024], we use the following metrics to assess the quality of the retrieved context and the generated responses while using text-embedding-3-small as the embedding model and GPT-4o as the LLM.

1. **Context Recall (C-Rec):** Proportion of claims in the ground-truth answer that can be attributed to the information present in the retrieved context.
2. **Context Entity Recall (C-ERec):** Proportion of entities in the ground-truth answer that are present in the retrieved context.
3. **Answer Similarity (A-Sim):** Similarity between the generated response and the ground-truth answer in the embedding space.
4. **Answer Correctness (A-Corr):** A combination of answer similarity (defined above) and factual similarity, which is the F1-score between the claims in the ground-truth answer and those in the generated response.
5. **Answer Relevance (A-Rel):** Measures how easily the original question can be inferred from the generated response.

Ethics Consideration. The survey was conducted in compliance with ACM’s Publications Policy on Human Research. No personal data was collected, and all responses were anonymized. For the agriculture and news domain experiments, only publicly available or proprietary datasets were used, with no sensitive data involved, adhering to ethical research standards.

6 Experiments

6.1 Query answering

6.1.1 Question Generation

We generate a set of question/answer pairs using the RAGAS framework [RAGAS, 2024] to validate the factual accuracy of our proposed method. RAGAS prompts off-the-shelf LLM to generate questions of varying difficulty, each with the corresponding ground-truth answers and contexts. Specifically, we generate up to 100 unique questions from RAGAS focused on multi-hop reasoning abilities, which is commonly required in specialized domain tasks. Examples of these generated questions, along with their ground-truth answers, are provided in Appendix C.

6.1.2 Does OG-RAG help in retrieving useful contexts?

A context is deemed useful for a query if it provides sufficient information to derive the ground-truth response. We evaluate this using Context Recall and Context Entity Recall. Table 1 compares the performance of different retrieval

表2: 不同检索方法下不同大语言模型生成的答案质量。我们发现所有指标95%置信区间为 ≤ 0.05 , 因此在此未报告。符号 ‘-’ 表示计算在1天内未完成。

方法	大豆					小麦					News				
	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ER	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel
<i>Llama-3-8B</i>															
RAG	0.22	0.07	0.26	0.59	0.22	0.14	0.05	0.26	0.65	0.23	0.01	0.01	0.15	0.52	0.08
RAPTOR	0.56	0.20	0.34	0.66	0.59	0.84	0.35	0.54	0.76	0.67	0.82	0.47	0.53	0.74	0.68
GraphRAG	0.46	0.11	0.26	0.63	0.52	-	-	0.43	0.35	0.27	-	-	-	-	-
OG-RAG	0.82	0.40	0.40	0.65	0.60	0.95	0.33	0.54	0.73	0.72	0.81	0.51	0.52	0.76	0.69
<i>Llama-3-70B</i>															
RAG	0.24	0.06	0.27	0.59	0.19	0.14	0.03	0.26	0.65	0.14	0.01	0.01	0.17	0.58	0.09
RAPTOR	0.55	0.23	0.41	0.70	0.64	0.85	0.39	0.58	0.77	0.75	0.82	0.47	0.39	0.72	0.64
GraphRAG	-	-	0.30	0.65	0.55	-	-	0.47	0.37	0.29	-	-	-	-	-
OG-RAG	0.84	0.41	0.54	0.75	0.56	0.95	0.31	0.63	0.77	0.73	0.70	0.69	0.51	0.77	0.67
<i>GPT-4o-mini</i>															
RAG	0.24	0.07	0.29	0.66	0.59	0.14	0.05	0.33	0.73	0.66	0.01	0.01	0.34	0.73	0.64
RAPTOR	0.59	0.23	0.34	0.68	0.85	0.84	0.36	0.51	0.77	0.88	0.842995	0.364377	0.51	0.77	0.88
GraphRAG	0.42	0.13	0.25	0.63	0.65	0.78	0.05	0.35	0.70	0.85	-	-	-	-	-
OG-RAG	0.83	0.41	0.48	0.72	0.77	0.95	0.33	0.62	0.78	0.85	0.81	0.51	0.62	0.78	0.85
<i>GPT-4o</i>															
RAG	0.22	0.08	0.31	0.62	0.29	0.14	0.04	0.29	0.69	0.28	0.01	0.01	0.27	0.67	0.20
RAPTOR	0.54	0.19	0.34	0.68	0.68	0.85	0.29	0.59	0.79	0.89	0.83	0.46	0.58	0.84	0.76
GraphRAG	0.41	0.14	0.26	0.63	0.63	0.82	0.52	0.35	0.70	0.86	-	-	-	-	-
OG-RAG	0.84	0.41	0.48	0.72	0.79	0.95	0.34	0.62	0.79	0.79	0.82	0.52	0.66	0.86	0.73

OG-RAG在几乎所有情况下都优于基线方法, 将正确声明的召回率提高了55%, 正确实体的召回率提高了110%。唯一的例外是新闻数据集, OG-RAG的上下文召回率与RAPTOR持平, 但仍表现更好。

6.1.3 OG-RAG 是否有助于生成事实准确性的回复?

一个有用的上下文在作为查询的一部分被添加到各种大语言模型中时, 应能引导生成更符合事实和精确的响应。我们通过比较检索上下文在不同大语言模型中添加时, 生成的响应/答案与真实答案的吻合程度来评估这一点。表2展示了3个数据集在响应正确性、相似性和相关性方面的结果。OG-RAG始终优于基线方法, 将答案正确性显著提高了40%, 答案相关性提高了16%。唯一值得注意的例外是, 在GPT-4o和Llama-3-70B的Wheat和Soybean数据集的答案相关性方面, OG-RAG表现略差。这可能是由于检索上下文的范围较广, 有时会引入不必要的信息。这可以通过进一步微调超图检索机制来缓解, 调整细节级别以适应预期查询的复杂性。我们将特定领域的优化留作未来工作, 因为当前方法已经在所有数据集上都提供了良好的响应。

6.1.4 IsOG-RAG高效吗?

最后, 我们通过比较OG-RAG在不同数据集上的预处理和每查询检索时间与其他方法的对比, 证明了其计算效率。表3显示, OG-RAG的效率几乎与简单的RAG方法相当, 尽管在事实准确性上至少提高了100%, 但在查询时间上仅增加了最多2秒。OG-RAG在预处理和查询阶段也显示出比RAPTOR和GraphRAG等更具竞争力的基线方法显著更低的计算时间, 尤其是在预处理时间上减少了50%, 这一点尤为突出。这种效率对于实时应用至关重要, 例如农业监测系统、法律研究和自动化新闻事实核查, 在这些应用中, 快速检索和处理领域特定知识是必不可少的。

6.2 上下文归因

6.2.1 调查设计

为了评估所提出的方法在帮助人类验证LLM生成响应中的事实方面的有效性, 我们进行了一项人类研究, 测量验证给定上下文是否支持生成响应所需的时间。我们从农业数据集中随机选择了10个查询, 并展示了由GPT-4o使用RAG和OG-RAG生成的响应, 每对响应都附带其各自的上下文。由于RAPTOR与RAG的内容相似性, 我们排除了RAPTOR; 由于GraphRAG的上下文长度过高, 我们也排除了GraphRAG。参与者被要求评估实际支持的程度

Table 2: Quality of the answers generated by different LLMs using different retrieval methods. We found the 95% confidence interval to be ≤ 0.05 for all metrics, so it is not reported here. The symbol ‘-’ denotes that the computation did not complete within 1 day.

Method	Soybean					Wheat					News				
	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel	C-Rec	C-ER	A-Corr	A-Sim	A-Rel	C-Rec	C-ERec	A-Corr	A-Sim	A-Rel
<i>Llama-3-8B</i>															
RAG	0.22	0.07	0.26	0.59	0.22	0.14	0.05	0.26	0.65	0.23	0.01	0.01	0.15	0.52	0.08
RAPTOR	0.56	0.20	0.34	0.66	0.59	0.84	0.35	0.54	0.76	0.67	0.82	0.47	0.53	0.74	0.68
GraphRAG	0.46	0.11	0.26	0.63	0.52	-	-	0.43	0.35	0.27	-	-	-	-	-
OG-RAG	0.82	0.40	0.40	0.65	0.60	0.95	0.33	0.54	0.73	0.72	0.81	0.51	0.52	0.76	0.69
<i>Llama-3-70B</i>															
RAG	0.24	0.06	0.27	0.59	0.19	0.14	0.03	0.26	0.65	0.14	0.01	0.01	0.17	0.58	0.09
RAPTOR	0.55	0.23	0.41	0.70	0.64	0.85	0.39	0.58	0.77	0.75	0.82	0.47	0.39	0.72	0.64
GraphRAG	-	-	0.30	0.65	0.55	-	-	0.47	0.37	0.29	-	-	-	-	-
OG-RAG	0.84	0.41	0.54	0.75	0.56	0.95	0.31	0.63	0.77	0.73	0.70	0.69	0.51	0.77	0.67
<i>GPT-4o-mini</i>															
RAG	0.24	0.07	0.29	0.66	0.59	0.14	0.05	0.33	0.73	0.66	0.01	0.01	0.34	0.73	0.64
RAPTOR	0.59	0.23	0.34	0.68	0.85	0.84	0.36	0.51	0.77	0.88	0.842995	0.364377	0.51	0.77	0.88
GraphRAG	0.42	0.13	0.25	0.63	0.65	0.78	0.05	0.35	0.70	0.85	-	-	-	-	-
OG-RAG	0.83	0.41	0.48	0.72	0.77	0.95	0.33	0.62	0.78	0.85	0.81	0.51	0.62	0.78	0.85
<i>GPT-4o</i>															
RAG	0.22	0.08	0.31	0.62	0.29	0.14	0.04	0.29	0.69	0.28	0.01	0.01	0.27	0.67	0.20
RAPTOR	0.54	0.19	0.34	0.68	0.68	0.85	0.29	0.59	0.79	0.89	0.83	0.46	0.58	0.84	0.76
GraphRAG	0.41	0.14	0.26	0.63	0.63	0.82	0.52	0.35	0.70	0.86	-	-	-	-	-
OG-RAG	0.84	0.41	0.48	0.72	0.79	0.95	0.34	0.62	0.79	0.79	0.82	0.52	0.66	0.86	0.73

methods across three datasets. OG-RAG outperforms the baselines in almost all cases, boosting the recall of correct claims by 55% and recall of correct entities by 110%. The only exception is the News dataset where OG-RAG matches the context recall performance of RAPTOR but still delivers better performance.

6.1.3 Does OG-RAG help generate factually accurate responses?

A useful context should lead to more factual and precise response when incorporated into the query for various LLMs. We evaluate this by comparing how closely the generated responses/answers align with the ground-truth answer when added as context across different LLMs. Table 2 presents the results of response correctness, similarity, and relevance for the 3 datasets. OG-RAG consistently outperforms the baselines, significantly improving answer correctness by 40%, and answer relevance by 16%. The only notable exception where OG-RAG slightly underperforms is in the Answer Relevance for Wheat and Soybean datasets in GPT-4o and Llama-3-70B. This is likely due to the broad scope of the retrieved context, which can sometimes introduce extraneous information. This can be possibly mitigated through further fine-tuning of the hypergraph retrieval mechanism, adjusting the level of detail to suit the complexity of the queries expected. We leave domain-specific optimization for future work, as the current approach already delivers good responses across all datasets.

6.1.4 Is OG-RAG efficient?

Finally, we demonstrate that OG-RAG is computationally efficient by comparing its pre-processing and per-query retrieval times with other methods across different datasets. Table 3 shows that OG-RAG performs nearly as efficiently as a simple RAG method, with only a minimal increase of at most 2 seconds during querying time despite being at least 100% better in factual accuracy. OG-RAG is also shown to have significantly lower computational time than more competitive baselines such as RAPTOR and GraphRAG at both the pre-processing and query stages, particularly highlighted by a 50% drop in the pre-processing times. This efficiency is particularly critical for real-time applications, such as agricultural monitoring systems, legal research, and automated news fact-checking, where quick retrieval and processing of domain-specific knowledge is essential.

6.2 Context attribution

6.2.1 Survey design

To assess how effectively the proposed method aids humans in verifying facts within LLM-generated responses, we conduct a human study measuring the time taken to verify whether the given context supports the generated response. We randomly select 10 queries from the agriculture dataset and present the responses generated by GPT-4o using both RAG and OG-RAG, each paired with their respective contexts. We exclude RAPTOR due to its content similarity with RAG, and GraphRAG due to its prohibitive context length. Participants are asked to evaluate the level of factual support

表3: 不同检索方法在特定领域问答中的效率。 T_{pre} 和 T_{query} 表示平均预处理和查询时间（秒）。我们发现方差在5秒以内，因此在此未报告。

方法	大豆			小麦			News		
	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $
RAG	11.41	2.49		10.55	2.36		449.21	3.56	
RAPTOR	71.66	4.81		61.56	4.38		1513.57	5.45	
GraphRAG	157.04	5.95		307.37	5.65		>1 day	-	
OG-RAG	29.61	3.75		47.76	4.09		655.15	4.12	

表4: 人类将生成响应归因于RAG和OG-RAG生成的上下文时，比较其容易程度的表格，并呈现了95%的置信区间。

方法	耗时 $\downarrow$	支持 [1-5] $\uparrow$
RAG	61.15 $\pm$ 28.48	2.67 $\pm$ 0.30
OG-RAG	43.50 $\pm$ 18.08	3.46 $\pm$ 0.19

上下文为响应提供了1-5的评分标准。我们还追踪了每位参与者完成此任务所需的时间。每位参与者会看到10个问题，其中包含5个随机查询，每个查询都与RAG和OG-RAG的响应及其上下文以随机顺序配对。为确保公平性，每个查询会在所有参与者中呈现相同次数。调查设计的示例可以在附录D中找到。

6.2.2 结果

共有16名参与者参与调查，他们年龄在18-34岁之间，熟悉大语言模型。表4展示了参与者完成平均时间以及他们对上下文的支持程度。我们观察到，OG-RAG平均减少了28.8%所需时间，并将人类归因的支持度提高了29.6%。这些结果表明，OG-RAG不仅能够加快事实验证的速度，还提供了更稳健、清晰的上下文，使系统在上下文事实归因方面更加用户友好和可靠。

6.3 事实演绎

6.3.1 演绎事实

我们通过评估 OG-RAG 在大语言模型中增强演绎推理的能力来评估其生成新结论的能力，这些结论基于一组预定义的事实。这些事实基于领域特定本体，为需要多步逻辑的推理任务提供框架。具体来说，在这个实验中，我们使用六个农业事实来推断 CO2 排放，因为这些信息在文档中不可直接获取。这些事实部分来源于关于化石燃料、农药和温室气体之间关系的行业资料。⁴:

1. 东北丘陵地带的农田面积为 1 公顷或 ha。
2. 北平原丘陵地带的农田面积为 2 公顷或 ha。
3. 除草剂产量通过将农田面积乘以推荐除草剂数量来计算。
4. 1 公斤的除草剂产量会导致 18.22–26.63 公斤的 CO2e 排放。
5. 1 公斤的杀虫剂产量会导致 14.79–18.91 公斤的 CO2e 排放。
6. 1 公斤的杀菌剂产量会导致 11.94–29.19 公斤的 CO2e 排放。

6.3.2 问题生成

为创建评估测试集，我们遵循 RAGAS 指南 RAGAS [2024]，提示 GPT-4o 生成需要应用演绎性事实以及从本体映射数据中随机采样片段来生成响应的问题。具体来说，我们使用以下提示：

⁴部分改编自 <https://www.panna.org/news/linking-fossil-fuels-and-pesticides-to-greenhouse-gases>

Table 3: Efficiency of different retrieval methods on domain-specific query-answering. T_{pre} and T_{query} denote the average pre-processing and query time in seconds. We found the variance to be within 5 seconds, so it is not reported here.

Method	Soybean			Wheat			News		
	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $	$T_{\text{pre}} \downarrow$	$T_{\text{query}} \downarrow$	$ \mathcal{C} $
RAG	11.41	2.49		10.55	2.36		449.21	3.56	
RAPTOR	71.66	4.81		61.56	4.38		1513.57	5.45	
GraphRAG	157.04	5.95		307.37	5.65		>1 day	-	
OG-RAG	29.61	3.75		47.76	4.09		655.15	4.12	

Table 4: Comparison of the ease with which humans can attribute generated responses to the contexts produced by RAG and OG-RAG, presented with 95% confidence intervals.

Method	Time taken $\downarrow$	Support [1-5] $\uparrow$
RAG	61.15 $\pm$ 28.48	2.67 $\pm$ 0.30
OG-RAG	43.50 $\pm$ 18.08	3.46 $\pm$ 0.19

the context provides for the response on a scale of 1-5. We also track the time each participant takes to complete this task. Each participant is shown 10 questions, consisting of 5 random queries, each paired with both RAG and OG-RAG responses and contexts in a randomized order. To ensure fairness, each query is presented an equal number of times across all participants. Examples of the survey design can be found in Appendix D.

6.2.2 Results

A total of 16 participants, aged 18-34 and familiar with LLMs, took part in the survey. Table 4 presents the average time taken and the level of support participants attributed to the contexts. We observed that OG-RAG significantly reduced the time required by 28.8% and increased the human-attributed support by 29.6% on average. These results demonstrate that OG-RAG not only enables faster fact verification but also provides more robust and clear contexts, making the system more user-friendly and reliable for context fact attribution.

6.3 Factual Deduction

6.3.1 Deductive Facts

We assess OG-RAG’s ability to enhance deductive reasoning in LLMs by evaluating how well it can generate new conclusions based on a set of predefined facts. These facts, grounded in domain-specific ontologies, provide the framework for reasoning tasks that require multi-step logic. Specifically, for this experiment we use six agricultural facts to deduce CO2 emissions, as this information is not directly available in the documents. These facts are partially derived from industry sources on the relationship between fossil fuels, pesticides, and greenhouse gases.⁴:

1. Farm area in the North Eastern Hill zone is 1 hectare or ha.
2. Farm area in North Plain Hill zone is 2 hectares or ha.
3. Herbicide production is calculated by multiplying the farm area by the recommended herbicide quantity.
4. 1 kg of herbicide production results in 18.22–26.63 kg of CO2e emissions.
5. 1 kg of insecticide production results in 14.79–18.91 kg of CO2e emissions.
6. 1 kg of fungicide production results in 11.94–29.19 kg of CO2e emissions.

6.3.2 Question Generation

To create the evaluation test set, we prompt GPT-4o following the RAGAS guidelines RAGAS [2024] to generate questions that require the application of deductive facts and a randomly sampled chunk from the ontology-mapped data to generate the responses. Specifically, we use the following prompt:

⁴Adapted in part from <https://www.panna.org/news/linking-fossil-fuels-and-pesticides-to-greenhouse-gases>

表5: 不同检索方法在支持不同大语言模型的演绎推理能力方面的比较。

方法	大豆			小麦		
	A-Corr	A-Sim	A-Rel	A-Corr	A-Sim	A-Rel
<i>GPT-4o-mini</i>						
RAG	0.46	0.89	0.66	0.41	0.92	0.64
RAPTOR	0.42	0.89	0.81	0.50	0.92	0.74
GraphRAG	0.44	0.91	0.83	0.49	0.93	0.82
OG-RAG	0.50	0.92	0.75	0.53	0.94	0.83
<i>GPT-4o</i>						
RAG	0.44	0.90	0.56	0.42	0.92	0.54
RAPTOR	0.01	0.11	0.03	0.41	0.91	0.74
GraphRAG	0.48	0.92	0.84	0.44	0.90	0.73
OG-RAG	0.56	0.92	0.75	0.47	0.94	0.83

给定以下数据和一个演绎规则集，生成一个需要应用规则对数据进行推理才能得出答案的难题。数据：<领域特定数据>规则：<固定规则集>问题：

接下来，我们进行两次额外的LLM调用，以生成相应的答案并为1到10分进行评分，评估该问题如何测试数据应用规则以推导出答案。我们选择了评分达到7分及以上的10个问题。生成的全部问题列表见附录C。

6.3.3 结果

表5展示了在两个农业数据集上使用GPT-4o和GPT-4o-mini作为底层大语言模型进行事实演绎的结果。在所有情况下，除了两个例外，与基线方法相比，OG-RAG上下文显著提高了生成答案的正确性、相似性和相关性。这证明OG-RAG在从固定事实集中支持演绎推理方面更有效。一个例外是在大豆数据集的答案相关性上，这再次指向由于OG-RAG检索到更广泛的上下文而导致答案稍显不相关。总体而言，这些结果证实，与替代检索方法相比，OG-RAG为演绎新事实提供了更鲁棒的上下文。

7 结论

在这项工作中，我们研究了使用本体引导检索增强生成技术对大语言模型进行领域自适应的问题。我们介绍了OG-RAG，这是一种基于超图的检索方法，通过使用特定领域的本体将文档中的事实结构化为超图，从而从文档中检索与查询相关的上下文。OG-RAG在包括医疗保健、法律和农业部门等工业 workflow 领域具有广泛适用性，以及新闻业、调查研究、咨询等知识驱动型任务。通过在两个农业数据集和一个新闻数据集上进行的大量实验，我们证明了OG-RAG显著提高了LLM生成响应的事实准确性，同时还能更快地将答案归因于其支持性上下文，并更有效地从领域事实中推导出结论。我们建议大语言模型应通过固定本体拥有更好的方式来整合受控词汇和结构化证据检索，因为这不仅增强了用户对生成响应的理解，还促进了大语言模型更顺畅地融入工业工作流和知识工作。通过提供更大的灵活性和对上下文检索和利用的控制，OG-RAG为更灵活可靠的语言系统铺平了道路。对于未来的工作，我们鼓励探索自动或半自动的本体构建技术，以端到端的方式构建这些框架，确保检索增强模型在各个不同领域具有更广泛的适用性。

Table 5: Comparison of different retrieval methods in their ability to support deductive reasoning from different LLMs.

Method	Soybean			Wheat		
	A-Corr	A-Sim	A-Rel	A-Corr	A-Sim	A-Rel
<i>GPT-4o-mini</i>						
RAG	0.46	0.89	0.66	0.41	0.92	0.64
RAPTOR	0.42	0.89	0.81	0.50	0.92	0.74
GraphRAG	0.44	0.91	0.83	0.49	0.93	0.82
OG-RAG	0.50	0.92	0.75	0.53	0.94	0.83
<i>GPT-4o</i>						
RAG	0.44	0.90	0.56	0.42	0.92	0.54
RAPTOR	0.01	0.11	0.03	0.41	0.91	0.74
GraphRAG	0.48	0.92	0.84	0.44	0.90	0.73
OG-RAG	0.56	0.92	0.75	0.47	0.94	0.83

Given the following data and a set of deductive rules, generate a hard question that requires the application of the rules on the data to generate the answer.
Data: < Domain-specific data >
Rules: < Fixed set of rules >
Question:

Next, we make two additional LLM calls to generate the corresponding answer and assign a rating from 1 to 10, evaluating how well the question tests the application of the rules on the data to derive the answer. We select 10 questions that receive a rating of at least 7. A full list of generated questions is provided in Appendix C.

6.3.3 Results

Table 5 presents the results of factual deductions across two agriculture datasets, using GPT-4o and GPT-4o-mini as the underlying LLMs. In all cases, except two, the OG-RAG context substantially improves the correctness, similarity, and relevance of the generated answers compared to baseline methods. This demonstrates that OG-RAG is more effective at supporting deductive reasoning from a fixed set of facts. One exception is in the Soybean dataset for answer relevance which again points to a slightly less pertinent answer due to a broader retrieved context by OG-RAG. Overall, these results confirm that OG-RAG provides a more robust context for deducing new facts than alternative retrieval methods.

7 Conclusion

In this work, we study the problem of domain adaptation of LLMs using ontology-grounded retrieval-augmented generation. We introduce OG-RAG, a novel hypergraph-based retrieval method that retrieves query-relevant context from documents by structuring their facts as a hypergraph using a domain-specific ontology. OG-RAG has wide applicability in domains which include industrial workflows in healthcare, legal, and agricultural sectors, among others as well as knowledge-driven tasks like news journalism, investigative research, consulting, and more. Through extensive experiments on two agriculture datasets and a news dataset, we demonstrate that OG-RAG significantly improves the factual accuracy of LLM-generated responses, while also enabling faster attribution of answers to their supporting context and more effectively deducing conclusions from domain facts. We recommend that LLMs have better ways to incorporate controlled vocabulary and structured evidence retrieval through fixed ontologies, as this not only enhances user comprehension of generated responses but also facilitates smoother integration of LLMs into industrial workflows and knowledge work. By offering greater flexibility and control over how context is retrieved and utilized, OG-RAG paves the way for more adaptable and reliable language systems. For future work we encourage to explore automated or

以构建这些框架，确保检索增强模型在各个不同领域具有更广泛的适用性。

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附录

数据集示例

A.1 大豆

标题：大豆：简介六大大豆生长区中的州分类：大豆主要种植在以下地区 — # 大豆：简介六大大豆生长区中的州分类：大豆主要种植在以下地区 ## 耕作

- 夏季收获后，必须进行深耕。这有助于将休眠昆虫暴露在极端高温和捕食性鸟类中，同时也有助于养分移动和土壤水分渗透。因此，每3-4年进行一次深耕，或者夏季进行一次常规深耕，然后进行2次十字耙或耕作以打破土块，将大豆的良好栽培提供理想的苗床。建议进行大豆栽培。此外，每4-5年进行一次深松作业，间隔10米，打破底土的紧实度，也有助于雨水渗透，这对即使在干旱期间也能实现作物持续生长非常有用。

A.2 小麦

- ## 气候
- 小麦作物具有广泛的适应性。它主要是一种温带作物，但在亚热带地区广泛种植，甚至在某些热带国家也有种植。
- 小麦种子的理想发芽温度约为20-23°C；尽管它们可以在3.5至35°C的温度范围内发芽。
 - 在抽穗和开花阶段，过高的温度或过低温度以及干旱对小麦有害。
 - 灌浆和发育时期的温度条件对产量非常关键。
 - 灌浆和发育期温度高于25°C时，往往会降低籽粒重量。当温度过高时，植物通过蒸腾作用会损失过多能量，剩余能量减少导致籽粒形成不良和产量降低。

A.3 新闻

雷霆队对阵雄狮队实时比分，赛况更新，来自NFL ‘周一晚间橄榄球’ 比赛的精华

作者: Dan Treacy **来源**: Sporting News **发布_于**: 2023-10-30T22:20:03+00:00 **分类**: 体育 **链接**: <https://www.sportingnews.com/us/nfl/news/雷霆队雄狮队实时比分精华-周一晚间橄榄球/d022b1d62b18af8a70c516f4> 雄狮队只需要在巴尔的摩遭遇惨败后重新回到胜利之列，他们在周一晚上面对主场球迷做到了这一点。底特律以26-14的比分击败了雷霆队，这场球赛的胜负感比比分显示的更为悬殊。雄狮队彻底压制了雷霆队，在得分上以329码的优势领先拉斯维加斯，但底特律的关键失误让分差缩小。进攻端在第一半场难以将红区机会转化为进球，而两场发生在雷霆队地盘的错误——包括Jared Goff的接球触地得分——让拉斯维加斯在下半场仍有机会竞争。更多信息：关于雄狮队复古 ‘檀香山蓝色’ 头盔的了解

Appendix

A Dataset Examples

A.1 Soybean

- title: SOYBEAN: AN INTRODUCTION Classification of States in six major Soybean growing zones: Soybean is majorly grown in the following areas —
- # SOYBEAN: AN INTRODUCTION Classification of States in six major Soybean growing zones: Soybean is majorly grown in the following areas
- ## TILLAGE
- Deep ploughing is essential during summer, after harvesting the Rabi crop. This facilitates exposing the hibernating insects to extreme heat and predatory birds as well as movement of nutrients and infiltration of soil water. Therefore, one deep ploughing once in 3-4 years, otherwise one normal ploughing in summer followed by 2 criss-cross harrowing or cultivation for breaking of soil clods will make ideal seed bed for a good crop of soybean cultivation is recommended. Also, sub-soiling operation once in 4-5 years at an interval of 10 meter, break the compactness of the sub-soil and also facilitate infiltration of rainwater which is useful for un-interrupted crop growth even during drought period also.

A.2 Wheat

- ## CLIMATE
- Wheat crop has wide adaptability. It is primarily a temperate crop but is widely cultivated in subtropical regions and is grown even in some tropical countries.
- Ideal temperature for germination of wheat seeds is around 20-23 o C though these can germinate in temperature range of 3.5 to 35 o C.
 - During the heading and flowering stages, excessively high or low temperatures and drought are harmful to wheat.
 - The temperature conditions at the time of grain filling and development are very crucial for yield.
 - Temperatures above 25 o C during grain filling and development period tend to depress grain weight. When temperatures are high, too much energy is lost through the process of transpiration by the plants and the reduced residual energy results in poorer grain formation and lower yields.

A.3 News

- # Raiders vs. Lions live score, updates, highlights from NFL ‘Monday Night Football’ game
- **author**: Dan Treacy
- **source**: Sporting News
- **published_at**: 2023-10-30T22:20:03+00:00
- **category**: sports
- **url**: <https://www.sportingnews.com/us/nfl/news/raiders-lions-live-score-highlights-monday-night-football/d022b1d62b18af8a70c516f4>
- The Lions just needed to get themselves back in the win column after a blowout loss in Baltimore, and they did just that in front of their home fans on Monday night. Detroit rolled to a 26-14 victory over the Raiders in a game that felt much more one-sided than the score indicates. The Lions thoroughly outplayed the Raiders, out-gaining Las Vegas by 329 yards, but critical mistakes by Detroit left points on the board. The offense struggled to turn red zone opportunities into touchdowns in the first half, and two turnovers deep in Raiders territory – including a pick-six by Jared Goff – kept Las Vegas in the game in the second half. MORE: What to know about Lions’ retro ‘Honolulu blue’ helmets

B 本体

B.1 本体提示

这里有一个关于小麦作物栽培本体的上下文定义。上下文定义：{context_定义}

使用以下数据以及上述作物栽培本体上下文定义，生成 JSON-LD。

在 JSON-LD 中使用 ’@graph’ 对象命名空间表示数据。要全面且确保填写所有数据。保持嵌套最小化，同时仍能消除歧义。如果一个 ’列表’ 命名空间中枚举了多个子字段，则不要将它们组合在一个子字段中，应保留为独立的子字段以消除歧义。确保您填充了 ’列表’ 命名空间中的所有项目，不要遗漏任何项目。不要在您的响应中包含任何解释或道歉。除了生成的 JSON-LD 之外，不要添加任何其他文本。以 JSON 格式生成。

数据：
{data} -----
JSON-LD json:

B.2 示例

图3和图4展示了工作中使用的农业和新闻本体。

C 评估问题

C.1 大豆

问题： 哪种害虫可以用 Imidacloprid 48 FS 农药在大豆作物中控制？ **OG-RAG 答案：** 可以在大豆作物中用 Imidacloprid 48 FS 农药控制的害虫是蚜虫。

问题： 什么 种植大豆应避免哪种土壤类型， 由于其中等至高的养分保持能力 和中等持水量 能力？ **RAPTOR Answer:** 上下文并未明确提到应避免哪种特定土壤来种植大豆，因为该土壤具有中等至较高的养分保蓄能力和中等的水分保蓄能力。然而， 它确实提供了针对不同土壤类型的建议：
- 在盐碱土（pH >7.5）中， 应施用150-200公斤/公顷的石膏，并配合有机肥。 - 在酸性土（pH <6.5）中， 建议施用600公斤/公顷的石灰。
根据这些信息，可以推断出虽然大豆可以在各种土壤类型中生长，但为了优化生长，盐碱土和酸性土需要特定的改良措施。因此， 并未明确建议避免特定的土壤类型，但针对不同土壤条件需要采取适当的土壤管理措施。
OG-RAG Answer: 应避免在沙质土中种植大豆，因为该土壤具有中等至较高的养分保蓄能力和中等的水分保蓄能力。

问题： 需要种子发芽率至少为80%、在85-90%的豆荚变褐并失去绿色时收获、并应储存在凉爽干燥处使用密封存储袋进行长期储存的作物的名称是什么？

B Ontology

B.1 Ontology Prompt

Here is a context definition for wheat crop cultivation ontology.
Context Definition:
{context_definition}

Generate a JSON-LD using the following data and the above context definition for crop cultivation ontology.
Use ’@graph’ object namespace for the data in JSON-LD.
Be comprehensive and make sure to fill all of the data.
Keep nesting to the minimum and still be able to disambiguate.
If there are multiple subfields enumerated in a ’List’ namespace then do not combine them in a single subfield, keep them as separate subfields to disambiguate.
Ensure that you populate all items in the ’List’ namespace, do not leave any item.
Do not include any explanations or apologies in your response.
Do not add any other text other than the generated JSON-LD in your response.
Generate in Json format.

Data:

{data}

JSON-LD json:

B.2 Examples

Figures 3 and 4 show agriculture and news ontologies used in the work.

C Evaluation Questions

C.1 Soybean

Question: Which pest can be controlled with Imidacloprid 48 FS pesticide in soybean crops?
OG-RAG Answer: The pest that can be controlled with Imidacloprid 48 FS pesticide in soybean crops is the White Grub.

Question: What type of soil should be avoided for growing soybeans due to its medium to high nutrient holding capacity and medium water holding capacity?
RAPTOR Answer: The context does not explicitly mention a specific type of soil that should be avoided for growing soybeans due to its medium to high nutrient holding capacity and medium water holding capacity. However, it does provide recommendations for managing different soil types:
- In saline soils (pH >7.5), gypsum at 150-200 kg/ha should be applied along with organic manure. - In acidic soils (pH <6.5), lime at 600 kg/ha is recommended.
Based on this information, it can be inferred that while soybeans can be grown in various soil types, specific amendments are needed for saline and acidic soils to optimize growth. Therefore, no particular soil type is explicitly advised to be avoided, but appropriate soil management practices are necessary for different soil conditions.
OG-RAG Answer: Sandy soil should be avoided for growing soybeans due to its medium to high nutrient holding capacity and medium water holding capacity.

Question: What is the name of the crop that requires seeds with a germination rate of at least 80%, should be harvested when 85-90% pods have turned brown and lost their green color, and should be stored in a cool, dry place using hermetic storage bags for long-term storage?

RAPTOR 回答: 提供的上下文并未提及需要种子发芽率至少为80%、在85-90%的豆荚变褐并失去绿色时收获、并应储存在凉爽干燥处使用密封存储袋进行长期储存的作物。上下文具体讨论了大豆的收获、脱粒和储存实践，包括发芽率至少为70%、在95%的豆荚变色时收获以及储存建议。因此，根据给定上下文，没有作物符合问题中描述的标准。

OG-RAG答案: 该作物的名称是大豆。

问题: 哪种大豆作物建议在中央区种植，并且早播期在2016年1月1日？

RAPTOR答案: 提供的上下文未提及任何建议在中央区种植且早播期在2016年1月1日的大豆作物。中央区的建议播种时间为6月20日至7月5日。因此，没有建议在中央区种植且播种时间为2016年1月1日的大豆作物。

OG-RAG答案: 建议在中央区种植且早播期从2016年1月1日开始的大豆作物是品种“JS 20-69”。

问题: 大豆发芽的温度范围是多少？它与开花和灌浆的温度要求相比如何？

RAPTOR Answer: 所提供的上下文并未明确说明大豆萌发、开花或灌浆所需的精确温度范围。因此，根据一般的农业知识：

- **大豆萌发**:** 大豆通常在15°C至30°C (59°F至86°F) 的土壤温度下萌发最佳。最佳萌发温度通常在25°C (77°F) 左右。
- **开花与灌浆**:** 对于开花和灌浆，大豆通常需要稍高的温度。这些阶段的最佳温度范围通常在20°C至30°C (68°F至86°F) 之间。温度高于30°C (86°F) 会对这些过程产生负面影响，尤其是当这种高温持续时间较长时。

总之，虽然大豆萌发可以在从约15°C开始的一系列温度下进行，但开花和灌浆的最佳条件通常处于一个相似但稍高的温度范围内，这强调了在整个生长期保持适度温度对实现最佳产量的重要性。

OG-RAG Answer: 大豆发芽的温度范围是15-30°C。对于开花，温度要求是20-30°C，而对于灌浆，它也是20-30°C。这意味着发芽的温度范围在较低端略宽于开花和灌浆，后两者具有相同的温度要求。

C.2 小麦

问题: 小麦收获的推荐含水量是多少？在储存前谷物应该如何干燥？

OG-RAG Answer: 收获小麦的推荐含水量为25%。在储存前，谷物应通过在阳光下摊铺在防水布塑料薄膜上将其干燥至含水量低于12%。

问题: 为什么在粮食生产方面要对小麦进行种子发芽试验？

OG-RAG Answer: 对小麦进行种子发芽试验对于确保最低发芽率为85%至关重要。这对于实现最佳作物密度至关重要，而作物密度直接影响产量。通过验证种子的发芽状态，农民可以确保他们种植的种子有可能成功生长，从而提高粮食生产和产量。

问题: 在 NEP 地区（西北平原区）储存前，收获和晾晒小麦籽粒的推荐做法是什么？

RAPTOR Answer: The context provided does not mention a crop that requires seeds with a germination rate of at least 80%, should be harvested when 85-90% pods have turned brown and lost their green color, and should be stored in a cool, dry place using hermetic storage bags for long-term storage. The context specifically discusses soybean harvesting, threshing, and storage practices, including a germination rate of at least 70%, harvesting when 95% of pods change color, and storage recommendations. Therefore, based on the given context, there is no crop that matches the criteria described in the question.
OG-RAG Answer: The name of the crop is Soybean.

Question: Which soybean crop is recommended for the Central Zone and has an early sowing time on January 1st, 2016?
RAPTOR Answer: The context provided does not mention any soybean crop recommended for the Central Zone with an early sowing time on January 1st, 2016. The recommended sowing times for the Central Zone are from 20th June to 5th July. Therefore, there is no soybean crop recommended for the Central Zone with a sowing time of January 1st, 2016.
OG-RAG Answer: The soybean crop recommended for the Central Zone with an early sowing time starting on January 1st, 2016, is the variety "JS 20-69."

Question: What is the temperature range for soybean germination and how does it compare to the temperature requirements for flowering and grain filling?
RAPTOR Answer: The provided context does not specify the exact temperature range required for soybean germination, flowering, or grain filling. Therefore, based on general agronomic knowledge:

- **Soybean Germination**:** Soybeans typically germinate best at soil temperatures between 15°C to 30°C (59°F to 86°F). Optimal germination usually occurs around 25°C (77°F).
- **Flowering and Grain Filling**:** For flowering and grain filling, soybeans generally require slightly higher temperatures. The optimal temperature range for these stages is typically between 20°C to 30°C (68°F to 86°F). Temperatures above 30°C (86°F) can negatively impact these processes, especially if they are sustained over long periods.

In summary, while soybean germination can occur at a range of temperatures starting from around 15°C, the optimal conditions for flowering and grain filling are generally within a similar but slightly higher range, emphasizing the importance of maintaining moderate temperatures throughout the growing season for optimal yield.
OG-RAG Answer: The temperature range for soybean germination is 15-30°C. For flowering, the temperature requirement is 20-30°C, and for grain filling, it is also 20-30°C. This means that the temperature range for germination is slightly broader on the lower end compared to flowering and grain filling, which have the same temperature requirements.

C.2 Wheat

Question: What is the recommended moisture level for harvesting wheat and how should the grain be dried before storage?
OG-RAG Answer: The recommended moisture level for harvesting wheat is 25%. Before storage, the grain should be dried by spreading it on tarpaulin plastic sheets in bright sunlight to a moisture level below 12

Question: Why conduct a seed germination test for wheat in relation to food production?
OG-RAG Answer: Conducting a seed germination test for wheat is essential to ensure a minimum germination rate of 85%. This is crucial for achieving an optimum plant population, which directly impacts the yield. By verifying the germination status of seeds, farmers can ensure that they plant seeds that are likely to grow successfully, leading to better food production and higher yields.

Question: What are the recommended practices for harvesting and drying wheat grains in the NEP Zone before storage?

OG-RAG 回答：在 NEP 地区（西北平原区）储存前，收获和晾晒小麦籽粒的推荐做法如下：

1. ****收割：**** - 当谷物含水量为25%时收割 - 小麦应在完全成熟前4-5天收割 - 早晨是收割的最佳时间 - 为快速收割，使用联合收割机以避免因破碎和倒伏造成的损失 - 人工收割时，捆成捆晾晒 3-4 天后再用脱粒机脱粒

2. ****干燥：**** - 储存前，将谷物摊铺在阳光充足的塑料布上晾晒 - 储存前确保含水量降至12%以下

问题：在北部山区，储存前推荐的小麦颗粒收割和干燥做法是什么？

OG-RAG答案：在北部山区，储存前推荐的小麦颗粒收割和干燥做法如下：

1. ****收割：**** - 当谷物含水量为25%时收割 - 小麦通常人工收割，但可以使用联合收割机快速收割以避免因破碎和倒伏造成的损失 - 在小麦完全成熟前4-5天收割 - 早晨是收割的最佳时间 - 人工收割时，捆成捆晾晒3-4天后再用脱粒机脱粒

2. ****干燥：**** - 存储前，将谷物摊铺在防水布塑料片上，在阳光下晾晒，以将含水量降至12%以下。

C.3 新闻

问题：哪些耳机时尚且透明？ **OG-RAG 回答：**时尚且透明的耳机是 “Nothing Ear Stick。”

Que问题：除了《飓风季节》，Netflix 11月1日还有哪些作品首播？
OG-RAG 回答：Netflix 11月1日，除了《飓风季节》，以下作品也将首播：

- 《Netflix 电影：困锁》 - 《信仰之谜》（Netflix纪录片） - 《新奥林匹斯》（Netflix电影） - 《至死方休：索林对海瑟姆》（Netflix纪录片） - 《空姐》（Netflix电影） - 《13岁到30岁》 - 《13小时：班加西秘密士兵》 - 《60天：第五季》 - 《阿达姆斯一家》 - 《大话西游》 - 《惊天交易》 - 《荒野求生》 - 《地狱归来》 - 《荒野之恋：第一季》 - 《玛黛亚的家庭团聚》 - 《木乃伊》（2017） - 《保罗·布莱特：商场保安》 - 《保罗·布莱特：商场保安2》 - 《完美音浪》 - 《斯卡布罗国王》 - 《斯科特·皮林对世界》 - 《小黄人》 - 《六尺之下：第1-5季》 - 《十六岁生日》 - 《社交网络》 - 《泰迪熊2》 - 《胜利之歌：第三季》 - 《辣手保姆》

问题：如何 price-match p政策抑制在生物 g 销售？
OG-RAG 回答：价格匹配政策通过确保客户无需在多家零售商处四处寻找就能获得最佳价格，帮助抑制在大型促销活动期间的客户流失。通过承诺匹配在其他地方找到的更低价格，像百思买、沃尔玛、戴尔、eBay 和劳氏这样的零售商旨在让客户满意并保持忠诚，防止他们从竞争对手处购买。这种策略在黑色星期五和网络星期一等主要促销活动期间尤其有效，当时价格竞争激烈。通过保证有竞争力的价格，零售商可以维持其客户群，并可能吸引那些寻找最佳交易的新客户。

问题：谁写了关于AI风险和创意阶层的内容？ **OG-RAG答案：**Daniel Tencer写了关于AI风险和创意阶层的内容。

OG-RAG Answer: The recommended practices for harvesting and drying wheat grains in the North Western Plains (NEP) Zone before storage are as follows:

1. ****Harvesting:**** - Harvest when the moisture level of the grain is 25- Wheat should be harvested 4-5 days before it is dead ripe. - Morning is the best time for harvesting. - For quick harvesting, use a combine harvester to avoid losses due to shattering and lodging. - In case of manual harvesting, make bundles and dry them for 3-4 days before threshing with a thresher.

2. ****Drying:**** - Before storage, dry the grain by spreading it on tarpaulin plastic sheets in bright sunlight. - Ensure the moisture level is reduced to below 12% before storage.

Question: What are the recommended practices for harvesting and drying wheat grains in the Northern Hills Zone before storage?
OG-RAG Answer: In the Northern Hills Zone, the recommended practices for harvesting and drying wheat grains before storage are as follows:

1. ****Harvesting:**** - Harvest when the moisture level of the grain is 25- Wheat is generally manually harvested, but a combine harvester can be used for quick harvesting to avoid losses due to shattering and lodging. - Harvest 4-5 days before the wheat is dead ripe. - Morning is the best time for harvesting. - In case of manual harvesting, make bundles and dry them for 3-4 days before threshing with a thresher.

2. ****Drying:**** - Before storage, dry the grain by spreading it on tarpaulin plastic sheets in bright sunlight to reduce the moisture level below 12%.

C.3 News

Question: Which earbuds are stylish and see-through?
OG-RAG Answer: The earbuds that are stylish and see-through are the "Nothing Ear Stick."

Question: What else debuts with 'Hurricane Season' on Netflix Nov. 1?
OG-RAG Answer: On Netflix Nov. 1, along with "Hurricane Season," the following titles debut:

- Locked In (Netflix Film) - Mysteries of the Faith (Netflix Documentary) - Nuovo Olimpo (Netflix Film) - Till Murder Do Us Part: Soering vs. Haysom (Netflix Documentary) - Wingwomen (Netflix Film) - 13 Going on 30 - 13 Hours: The Secret Soldiers of Benghazi - 60 Days In: Season 5 - The Addams Family - The Big Lebowski - The Change-Up - Desperado - Downsizing - Drag Me to Hell - Love in the Wild: Season 1 - Madea’s Family Reunion - The Mummy (2017) - Paul Blart: Mall Cop - Paul Blart: Mall Cop 2 - Pitch Perfect - The Scorpion King - Scott Pilgrim vs. the World - Sherlock Gnomes - Six Feet Under: Seasons 1-5 - Sixteen Candles - The Social Network - Ted 2 - Victorious: Season 3 - Whiplash

Question: How do price-match policies curb customer loss in big sales?
OG-RAG Answer: Price-match policies help curb customer loss during big sales by ensuring that customers can get the best price without needing to shop around at multiple retailers. By offering to match lower prices found elsewhere, retailers like Best Buy, Walmart, Dell, eBay, and Lowe’s aim to keep customers satisfied and loyal, preventing them from purchasing from competitors. This strategy is particularly effective during major sales events like Black Friday and Cyber Monday, when price competition is intense. By guaranteeing competitive pricing, retailers can maintain their customer base and potentially attract new customers who are looking for the best deals.

Question: Who wrote about AI risks and the creative class?
OG-RAG Answer: Daniel Tencer wrote about AI risks and the creative class.

OG-RAG: 本体引导检索增强生成用于大语言模型预印本
问题： 谁在对阵泰潘斯的比赛中命中了老鹰队的制胜绝杀球？ OG-RAG 答案： 泰勒·哈维在对阵泰潘斯的比赛中命中了老鹰队的制胜绝杀球。
问题： Steven Tey开发的哪个Chrome扩展有助于分享AI回复？ OG-RAG答案： Steven Tey开发的有助于分享AI回复的Chrome扩展叫做"ShareGPT。"
问题： 哪些艺术家在Capitol Music Group的Michelle Jubelirer手下取得了成功？ OG-RAG答案： 在Capitol Music Group的Michelle Jubelirer手下取得成功的艺术家包括IceSpice和The Beatles。
问题： 为什么JLab耳机被视为带均衡器预设的预算友好型备用耳机？ OG-RAG答案： JLab耳机被视为预算友好型备用耳机，因为它们通过将三种不同的均衡器预设直接集成到耳机中提供了很高的性价比。这一功能在不增加高成本的情况下提升了听音体验，使其成为那些寻求主要无线耳机经济替代方案的人的吸引人选择。
C.4 事实演绎 问题： 计算位于东北平原区的一个农场使用推荐除草剂（针对Phalaris minor）生产除草剂所产生的总CO2e排放量。假设农场面积为1公顷。根据给定数据和规则，提供CO2e排放量的范围。 OG-RAG答案： 要计算位于东北平原区的一个农场使用推荐除草剂生产除草剂所产生的总CO2e排放量，我们需要遵循以下步骤： **确定推荐的除草剂和数量： ** - 推荐的除草剂是Clodinafop。 - 推荐的数量是每公顷60克有效成分（a.i。）。 **计算所需的总除草剂量： ** - 由于农场面积为1公顷，所需的总除草剂量为60克。 **将克转换为千克： ** - 60克相当于0.06千克。 **计算CO2e排放量： ** - 除草剂生产的CO2e排放量范围为每千克除草剂18.22至26.63千克的CO2e。 - 对于0.06千克的除草剂，CO2e排放量的计算如下： - 下限：0.06 kg * 18.22 kg CO2e/kg = 1.0932 kg CO2e - 上限：0.06 kg * 26.63 kg CO2e/kg =1.5978 kg CO2e 因此，该农场除草剂生产的总CO2e排放量在1.0932至1.5978 kg CO2e之间。
问题： 计算位于华北平原地区的农场，在推荐除草剂用量为每公顷2 kg、杀虫剂用量为每公顷1.5 kg、杀菌剂用量为每公顷1 kg的假设下，除草剂、杀虫剂和杀菌剂生产产生的总潜在CO2e排放量。使用平均CO2e值进行计算。 OG-RAG Answer: 要计算位于华北平原地区的农场，除草剂、杀虫剂和杀菌剂生产产生的总潜在CO2e排放量，因此，该农场除草剂生产的总CO2e排放量在1.0932至1.5978 kg CO2e之间。 **确定农田面积**： 北平原山区农田面积为2公顷。 **计算每种化学品的总使用量**： - 除草剂：2 公斤/公顷 × 2 公顷 = 4 公斤 - 杀虫剂：1.5 公斤/公顷 × 2 公顷 = 3 公斤 - 杀菌剂：1 公斤/公顷 × 2 公顷 = 2 公斤 **使用每种化学品的平均CO2e值**： - 除草剂：平均CO2e = (18.22 + 26.63) / 2 = 22.425 kg CO2e/kg - 杀虫剂：平均CO2e = (14.79 + 18.91) / 2 = 16.85 kg CO2e/kg - 杀菌剂：平均CO2e = (11.94 + 29.19) / 2 = 20.565 kg CO2e/kg

OG-RAG: Ontology-Grounded Retrieval-Augmented Generation For Large Language ModelsA PREPRINT
Question: Who hit the Hawks’ game-winning buzzer-beater vs. the Taipans? OG-RAG Answer: Tyler Harvey hit the Hawks’ game-winning buzzer-beater vs. the Taipans.
Question: Which Chrome extension by Steven Tey aids in sharing AI replies? OG-RAG Answer: The Chrome extension by Steven Tey that aids in sharing AI replies is called "ShareGPT."
Question: Which artists thrived under Michelle Jubelirer at Capitol Music Group? OG-RAG Answer: The artists that thrived under Michelle Jubelirer at Capitol Music Group include Ice Spice and The Beatles.
Question: Why are JLab earphones seen as a budget-friendly backup with EQ presets? OG-RAG Answer: JLab earphones are considered a budget-friendly backup because they offer great value by including three different EQ presets directly into the earphones. This feature enhances the listening experience without a high cost, making them an attractive option for those seeking an affordable alternative to their main wireless earphones.
C.4 Factual Deduction Question: Calculate the total CO2e emissions from herbicide production for a farm located in the North Eastern Plains Zone, using the recommended herbicide for Phalaris minor. Assume the farm area is 1 hectare. Provide the range of CO2e emissions based on the given data and rules. OG-RAG Answer: To calculate the total CO2e emissions from herbicide production for a farm located in the North Eastern Plains Zone, we need to follow these steps: **Identify the Recommended Herbicide and Quantity:** - The recommended herbicide for Phalaris minor is Clodinafop. - The recommended quantity is 60 gm a.i. (active ingredient) per hectare. **Calculate the Total Herbicide Required:** - Since the farm area is 1 hectare, the total herbicide required is 60 grams. **Convert Grams to Kilograms:** - 60 grams is equivalent to 0.06 kilograms. **Calculate CO2e Emissions:** - The CO2e emissions for herbicide production range from 18.22 to 26.63 kg of CO2e per kilogram of herbicide. - For 0.06 kg of herbicide, the CO2e emissions are calculated as follows: - Lower bound: 0.06 kg * 18.22 kg CO2e/kg = 1.0932 kg CO2e - Upper bound: 0.06 kg * 26.63 kg CO2e/kg = 1.5978 kg CO2e Therefore, the total CO2e emissions from herbicide production for the farm are in the range of 1.0932 to 1.5978 kg CO2e.
Question: Calculate the total potential CO2e emissions from herbicide, insecticide, and fungicide production for a farm located in the North Plain zone, assuming the recommended herbicide quantity is 2 kg per hectare, insecticide quantity is 1.5 kg per hectare, and fungicide quantity is 1 kg per hectare. Use the average CO2e values for your calculations. OG-RAG Answer: To calculate the total potential CO2e emissions from herbicide, insecticide, and fungicide production for a farm located in the North Plain zone, we need to follow these steps: **Determine the farm area**: The farm area in the North Plain Hill zone is 2 hectares. **Calculate the total quantity of each type of chemical used**: - Herbicide: 2 kg/ha × 2 ha = 4 kg - Insecticide: 1.5 kg/ha × 2 ha = 3 kg - Fungicide: 1 kg/ha × 2 ha = 2 kg **Use the average CO2e values for each type of chemical**: - Herbicide: Average CO2e = (18.22 + 26.63) / 2 = 22.425 kg CO2e/kg - Insecticide: Average CO2e = (14.79 + 18.91) / 2 = 16.85 kg CO2e/kg - Fungicide: Average CO2e = (11.94 + 29.19) / 2 = 20.565 kg CO2e/kg

4. **计算每种化学品的总CO2e排放量**： - 除草剂： $4\text{ kg} \times 22.425\text{ kg CO2e/kg} = 89.7\text{ kg CO2e}$ - 杀虫剂： $3\text{ kg} \times 16.85\text{ kg CO2e/kg} = 50.55\text{ kg CO2e}$ - 杀菌剂： $2\text{ kg} \times 20.565\text{ kg CO2e/kg} = 41.13\text{ kg CO2e}$

5. **计算总潜在CO2e排放量**： - 总CO2e = $89.7\text{ kg} + 50.55\text{ kg} + 41.13\text{ kg} = 181.38\text{ kg CO2e}$

因此，该农场在黄淮海平原区的除草剂、杀虫剂和杀菌剂生产中，总潜在CO2e排放量为181.38 kg CO2e。

问题： 计算该农场在东北丘陵区的除草剂生产中总CO2e排放量，假设推荐的除草剂用量为每公顷3公斤。使用每公斤除草剂CO2e排放量的范围来确定可能的最小和最大排放量。

OG-RAG答案： 要计算该农场在东北丘陵区的除草剂生产中总CO2e排放量，我们需要按照以下步骤进行计算：

1. **确定农场面积**： 根据上下文，东北丘陵区的农场面积为1公顷。

2. **计算除草剂总数量**： 推荐的除草剂数量是每公顷3公斤。因此，对于1公顷的农场，除草剂总数量是：

$$3\text{ kg/ha} \times 1\text{ ha} = 3\text{ kg}$$

3. **计算CO2e排放量**： 每公斤除草剂的CO2e排放量在18.22至26.63公斤CO2e之间。因此，3公斤除草剂的总CO2e排放量是：

- **最低排放量**：

$$3\text{ kg} \times 18.22\text{ kg CO2e/kg} = 54.66\text{ kg CO2e}$$

- **最高排放量**：

$$3\text{ kg} \times 26.63\text{ kg CO2e/kg} = 79.89\text{ kg CO2e}$$

因此，该农场在东北山区的除草剂生产总CO2e排放量在54.66公斤至79.89公斤CO2e之间。

D 调查设计

图5展示了我们调查的设计，包括引言和一个示例问题。

4. **Calculate the total CO2e emissions for each type of chemical**： - Herbicide: $4\text{ kg} \times 22.425\text{ kg CO2e/kg} = 89.7\text{ kg CO2e}$ - Insecticide: $3\text{ kg} \times 16.85\text{ kg CO2e/kg} = 50.55\text{ kg CO2e}$ - Fungicide: $2\text{ kg} \times 20.565\text{ kg CO2e/kg} = 41.13\text{ kg CO2e}$

5. **Calculate the total potential CO2e emissions**： - Total CO2e = $89.7\text{ kg} + 50.55\text{ kg} + 41.13\text{ kg} = 181.38\text{ kg CO2e}$

Therefore, the total potential CO2e emissions from herbicide, insecticide, and fungicide production for the farm in the North Plain zone is 181.38 kg CO2e.

Question: Calculate the total CO2e emissions from herbicide production for a farm in the North Eastern Hill zone, assuming the recommended herbicide quantity is 3 kg per hectare. Use the range provided for CO2e emissions per kg of herbicide to determine the minimum and maximum possible emissions.

OG-RAG Answer: To calculate the total CO2e emissions from herbicide production for a farm in the North Eastern Hill zone, we need to follow these steps:

1. **Determine the farm area**： According to the context, the farm area in the North Eastern Hill zone is 1 hectare.

2. **Calculate the total herbicide quantity**： The recommended herbicide quantity is 3 kg per hectare. Therefore, for a 1-hectare farm, the total herbicide quantity is:

$$3\text{ kg/ha} \times 1\text{ ha} = 3\text{ kg}$$

3. **Calculate the CO2e emissions**： The CO2e emissions per kg of herbicide range from 18.22 to 26.63 kg of CO2e. Therefore, the total CO2e emissions for 3 kg of herbicide are:

- **Minimum emissions**：

$$3\text{ kg} \times 18.22\text{ kg CO2e/kg} = 54.66\text{ kg CO2e}$$

- **Maximum emissions**：

$$3\text{ kg} \times 26.63\text{ kg CO2e/kg} = 79.89\text{ kg CO2e}$$

Therefore, the total CO2e emissions from herbicide production for the farm in the North Eastern Hill zone range from 54.66 kg to 79.89 kg of CO2e.

D Survey design

Figure 5 shows the design of our survey with the introduction and one exemplary question.

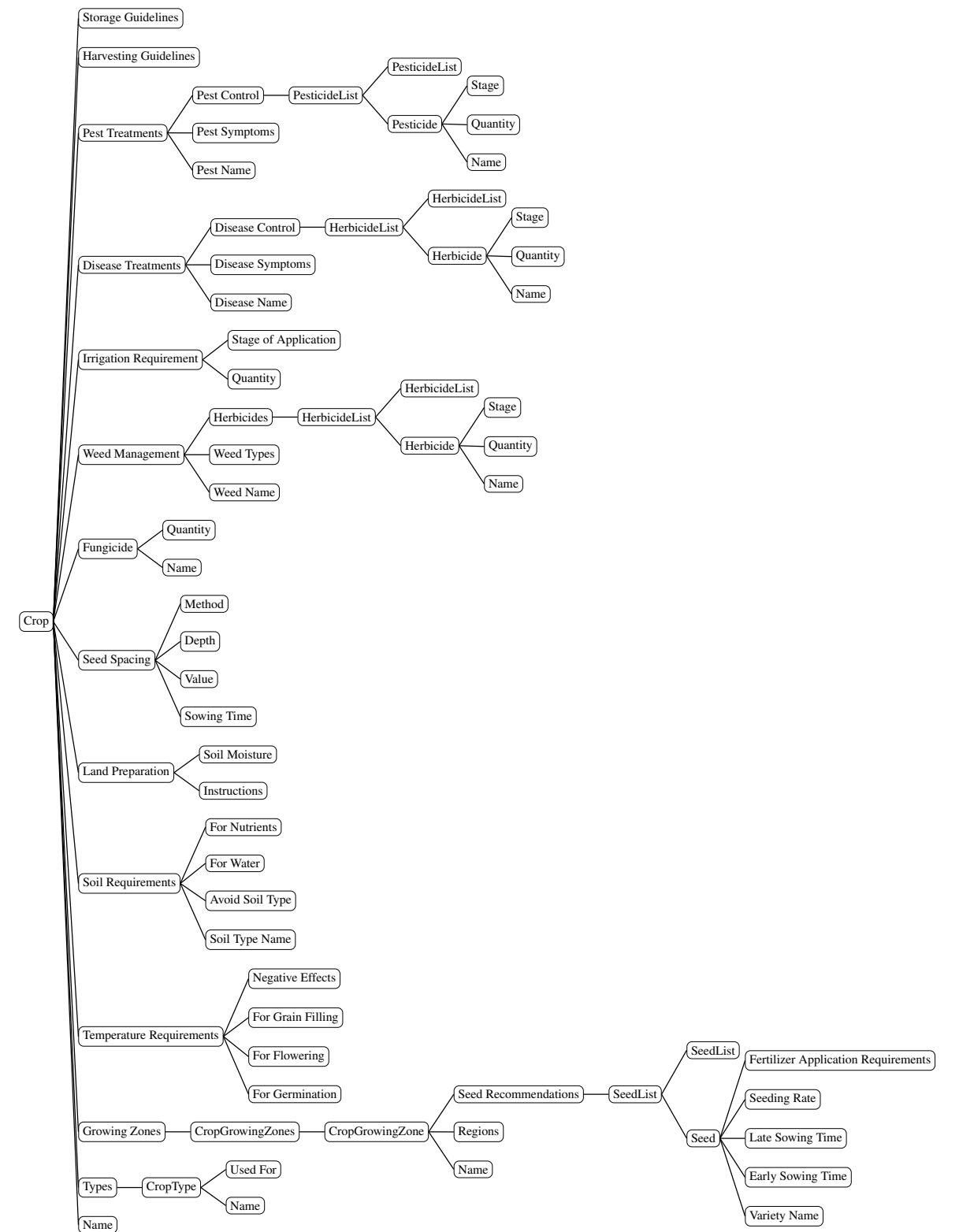


Figure 3: Agriculture ontology

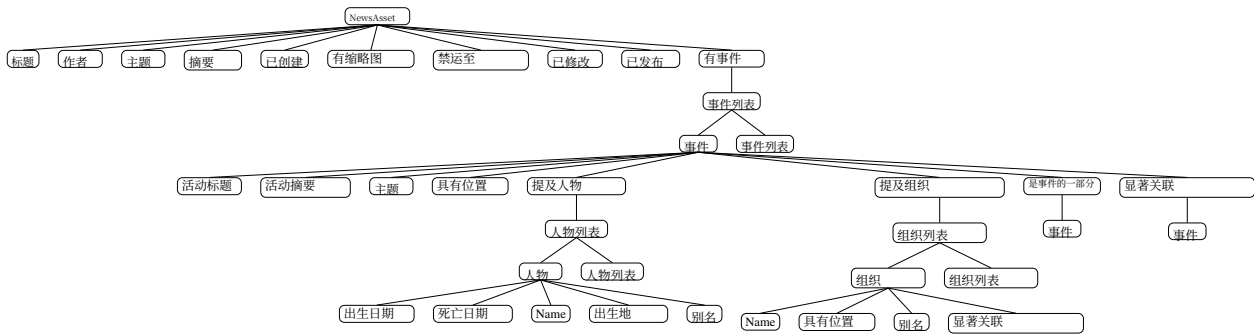


图4: 新闻本体论

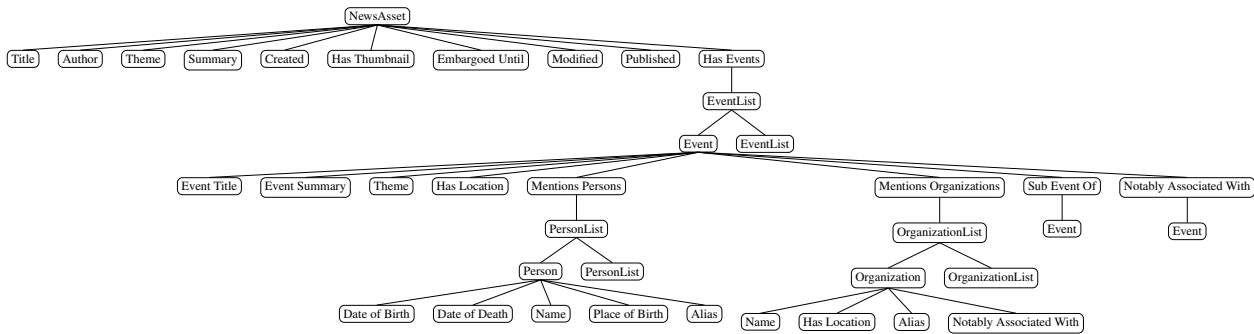


Figure 4: News ontology

你多大了?

18岁以下	☐
18-24岁	☐
25-34岁	☐
35-44岁	☐
45-54岁	☐
55-64岁	☐
65+ 岁	☐

你对语言模型有多熟悉?

完全不熟悉	☐
有点熟悉但不常用	☐
相当熟悉但不常用	☐
非常熟悉且经常使用	☐

(a) 介绍页面

根据下方查询，你需要使用附加的上下文验证响应中的声明。

查询: 需要使用发芽率至少为80%的种子的作物名称是什么? 该作物应在85-90%的豆类发芽率并失去绿色时收获, 并应使用密封存储袋保存在阴凉干燥处进行长期储存。

回答
大豆
上下文
收割与脱粒 • 大豆的收割最佳时机非常重要, 因为脱粒会导致产量损失。当田间风化会导致种子活力下降。 • 大豆豆荚颜色开始变化 (生理成熟) 是进行收割的正确指示。此时, 大豆种子的含水量约为 14-16%。 因此, 建议农民等待到叶片/植株变黄才收获。当90%的豆荚从绿色变为其他颜色时即可收获。 • 将1-2.5天, 收获的大豆作物可进行脱粒。如果要在后续作物播种后进行脱粒, 收获的大豆最好存放在阴凉干燥处, 并密封在密封袋中。建议将大豆存放在阴凉干燥处, 为了保持种子的活力并避免脱粒机械损坏, 脱粒应在320-400 rpm的速度下进行。 • 脱粒后的大豆应再次晾干 3-4天。脱水应降至10%, 这有助于防止霉菌和发芽率下降。 • 脱粒后应密封存储, 避免阳光直射。大豆应存放在阴凉干燥处, 如果受潮, 高度应不超过4-5磅, 且不超过5英尺 (使用平白), 以保持大豆种子的活力/发芽率。 • 在将种子储存在罐中时, 应小心放置在适当的通风/平台上。种子不应直接储存在罐中, 因为罐中的水分渗透可能是导致霉菌污染的一个来源。因此将重袋这种状况以进行存储。种子发芽试验

这个上下文对验证响应中的声明有多大用?

完全无用	☐
有用但不针对查询	☐
支持部分主张	☐
支持所有主张	☐

(b) RAG上下文归因

根据以下查询, 您需要使用提供的上下文验证响应中的声明。

查询: 需要种子发芽率至少为80%的作物的名称是什么? 该作物应在85-90%的豆类发芽率并失去绿色时收获, 并应使用密封存储袋保存在阴凉干燥的地方进行长期储存。

响应 (可滚动)	
大豆	
上下文 (两侧均为真)	
作物名称: 大豆 收获时间: 当85-90%的豆荚变黄并失去绿色时。 豆荚变黄并失去绿色时, 种子的含水量约为 10-12%。 种子存储要求: - 存储名称: 罐装豆 - 存储数量: 2-3 克/千克种子 - 存储湿度: 应具有至少 80% 的发芽率。 种子发芽试验要求: 种子应存放在阴凉干燥处。 长期储存应使用密封存储袋。	作物名称: 大豆 收获时间: 当85-90%的豆荚变黄并失去绿色时。 豆荚变黄并失去绿色时, 种子的含水量约为 10-12%。 种子存储要求: - 存储名称: 罐装豆 - 存储数量: 每公斤种子2克 - 存储湿度: 应具有超过80%的发芽率。 种子发芽试验要求: 种子应存放在阴凉干燥处。 长期储存应使用密封存储袋。

这个上下文对验证响应中的声明有多有用?

完全无用	☐
有用但不针对查询	☐
支持部分主张	☐
支持所有主张	☐

(c) OG-RAG上下文归因

图5: 调查设计

How old are you?

Under 18	☐
18-24 years old	☐
25-34 years old	☐
35-44 years old	☐
45-54 years old	☐
55-64 years old	☐
65+ years old	☐

How familiar are you with language models?

Not at all familiar	☐
Somewhat familiar but don't use them often	☐
Quite familiar but don't use them often	☐
Very familiar and use them often	☐

(a) Introduction page

Timing

These page timer metrics will not be displayed to the recipient.

First Click	4.763 seconds
Last Click	4.763 seconds
Page Submit	0 seconds
Click Count	1 clicks

Given the query below, you are supposed to verify the claims in the response using the adjoining context.

Query: What is the name of the crop that requires seeds with a germination rate of at least 80%, should be harvested when 85-90% pods have turned brown and lost their green color, and should be stored in a cool, dry place using hermetic storage bags for long-term storage?

Response
Soybean
Context
HARVESTING & THRESHING • Optimum time of harvesting is very important for soybean as it causes yield loss due to shattering and seed viability loss due to field weathering. • Initiation of change of pod color (physiological maturity) of soybean is the right indication to go for harvesting. At this time, the moisture percentage of soybean seed is around 14-16%. Hence, the farmers are advised not to wait till the foliage/plant turns yellow. Harvesting can be done when 90% pod changes color from green. • The harvested soybean crop, after sun-drying for 2-3 days is ready for threshing. Because of sowing of subsequent crop if threshing is to be performed later, the harvested soybean should be collected preferably on threshing floor and covered with tarpaulin to avoid damage from rain/shattering. To maintain viability of the seed and to avoid bio/mechanical damage, threshing should be done at 100-400 rpm. • The threshed soybean should again be allowed to sun-drying for 3-4 days in order to bring down the moisture up to 10% which is essential to avoid fungal infection during storage. • The storage place should be cool with airtight and moist free. The soybean bags should be kept upright as far as possible. If stacking is to be done, it should be only up to 4-5 bags of not more than 5 feet height using platform in order to maintain the viability germination of soybean seed. • While moving the seed bags to storage house, it should be carefully placed at the appropriate place/platform. The seed bags should not be in direct contact with floor/wall. The moisture seepage in the wall/floor may be a source of infection of diseases, hence can be avoided for storage. SEED GERMINATION TEST Farmers are advised to check germination status of seed purchased/available with them before sowing. To ensure optimum plant population and thereby good yield, minimum 70% germination is essential. This can be done through sowing of 100 seeds in 1mX1m plot and it is kept moist. From 3-8 days emergence is counted everyday till the count is fulfilled. The germination test can also be done by placing 100 seeds in between two newspaper sheets and rolling them with a moist cloth.

How useful is the context for verifying the claims of the response?

Not at all useful	☐
Can be useful but not specific to the query	☐
Provides support for some claims	☐
Provides support for all claims	☐

(b) RAG context attribution

Timing

These page timer metrics will not be displayed to the recipient.

First Click	0 seconds
Last Click	0 seconds
Page Submit	0 seconds
Click Count	0 clicks

Given the query below, you are supposed to verify the claims in the response using the adjoining context.

Query: What is the name of the crop that requires seeds with a germination rate of at least 80%, should be harvested when 85-90% pods have turned brown and lost their green color, and should be stored in a cool, dry place using hermetic storage bags for long-term storage?

Response (scrollable)	
Soybean	
Context (both sides are true)	
Crop Name: Soybean Harvesting Guidelines: Harvest when 85-90% pods have turned brown and lost their green color. The moisture content of the seeds should be around 10-12%. Seed Fungicide Requirements: - Fungicide Name: Thiram or Carbendazim Seed Germination Test Requirements: Seeds should have more than 80% germination. Storage Guidelines: Store in a cool and dry place. Use hermetic storage bags for long-term storage.	Crop Name: Soybean Harvesting Guidelines: Harvest when 85-90% pods turn brown and hard. Seed Fungicide Requirements: - Fungicide Name: Thiram or Carbendazim Seed Germination Test Requirements: Seeds should have more than 80% germination. Storage Guidelines: Store in a cool and dry place. Use hermetic storage bags for long-term storage.

How useful is the context for verifying the claims of the response?

Not at all useful	☐
Can be useful but not specific to the query	☐
Provides support for some claims	☐
Provides support for all claims	☐

(c) OG-RAG context attribution

Figure 5: Survey design